

Cross-Wearer Gesture Recognition — LOUO Variant Comparison

Generated 2026-05-14 06:58:45

Data & configuration

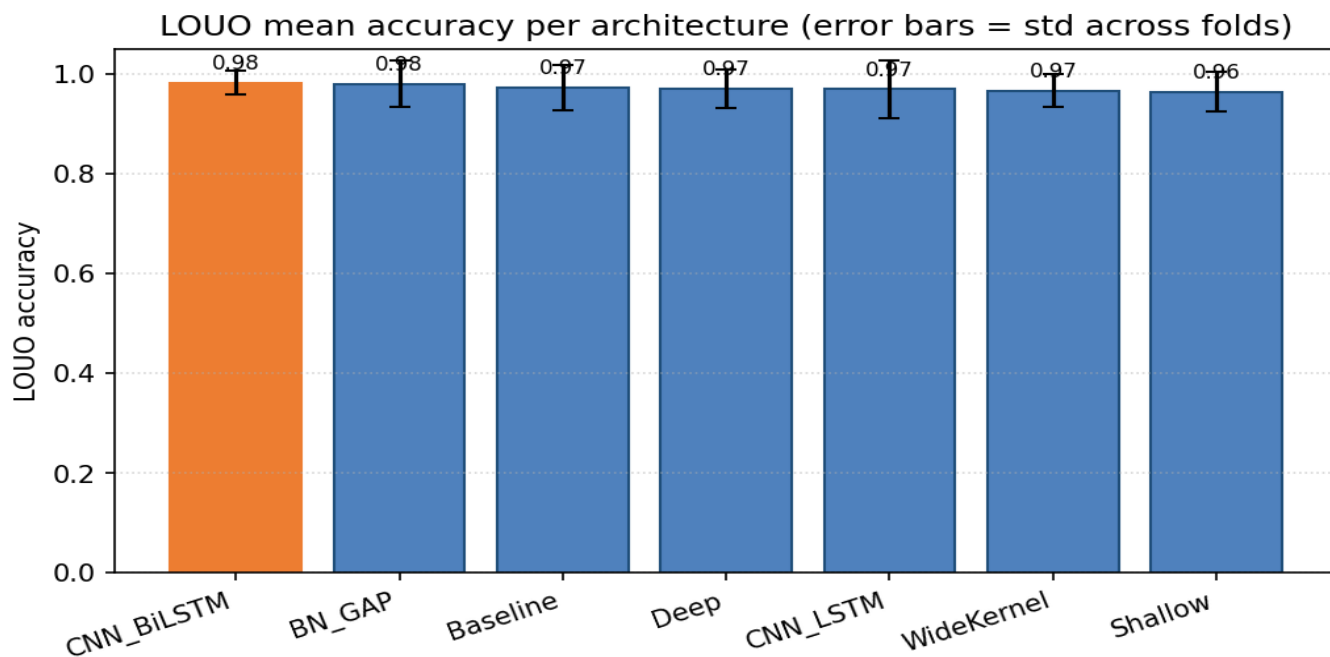
LOUO path 1	/home/jestin/ThesisRepo/ML/NewTestData/1_Alan/Dynamic
LOUO path 2	/home/jestin/ThesisRepo/ML/NewTestData/2_Alex/Dynamic
LOUO path 3	/home/jestin/ThesisRepo/ML/NewTestData/3_Anghad/Dynamic
LOUO path 4	/home/jestin/ThesisRepo/ML/NewTestData/4_Daniel/Dynamic
LOUO path 5	/home/jestin/ThesisRepo/ML/NewTestData/6_Harry/Dynamic
LOUO path 6	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Dynamic
LOUO path 7	/home/jestin/ThesisRepo/ML/NewTestData/11_Mansh/Dynamic
LOUO path 8	/home/jestin/ThesisRepo/ML/NewTestData/12_Marcus/Dynamic
LOUO path 9	/home/jestin/ThesisRepo/ML/NewTestData/14_StephenV2/Dynamic
LOUO path 10	/home/jestin/ThesisRepo/ML/NewTestData/15_Tash/Dynamic
LOUO path 11	/home/jestin/ThesisRepo/ML/NewTestData/16_Bella/Dynamic
LOUO path 12	/home/jestin/ThesisRepo/ML/NewTestData/17_Raquel/Dynamic
LOUO path 13	/home/jestin/ThesisRepo/ML/NewTestData/18_Bridgette/Dynamic
LOUO users loaded	13 (11_Mansh, 12_Marcus, 14_StephenV2, 15_Tash, 16_Bella, 17_Raquel, 18_Bridgette, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 6_Harry, 8_Jestin)
Excluded classes	Double_Nothing
Classes	6: Double_Lower, Double_Pistol_Recoil, Double_Raise, Double_Wiggle, Double_cmere, Drum_Roll
Sensor channels	176 (sequence length 30)
Resample / filter	30 steps · Butterworth cutoff 10.0 Hz, order 4
Per-trial normalisation	zscore
Downstream normalisation	minmax
Sampling rate	30.0 Hz
Random state	42
Augmentation	off
Epochs · batch size	40 · 16
Early stopping	on · monitor val_loss · patience=8
LOUO inner-validation	user
LOUO variants	Baseline, Shallow, Deep, BN_GAP, WideKernel, CNN_LSTM, CNN_BiLSTM
Final-retrain variant	BN_GAP_Wide
Final-retrain held-out user	— (trained on all users)

Variant comparison (LOUO)

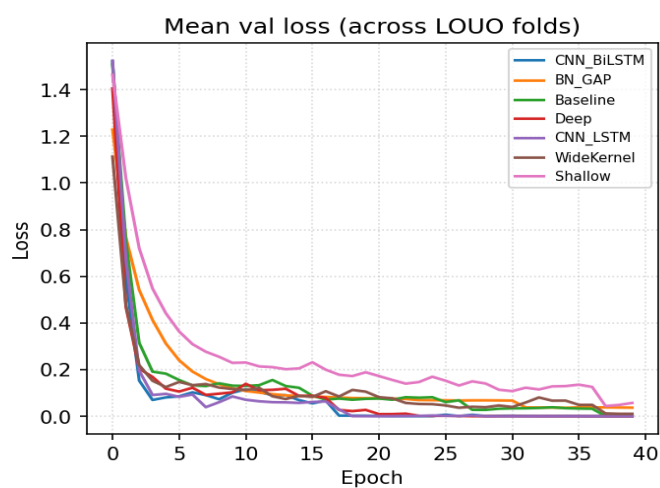
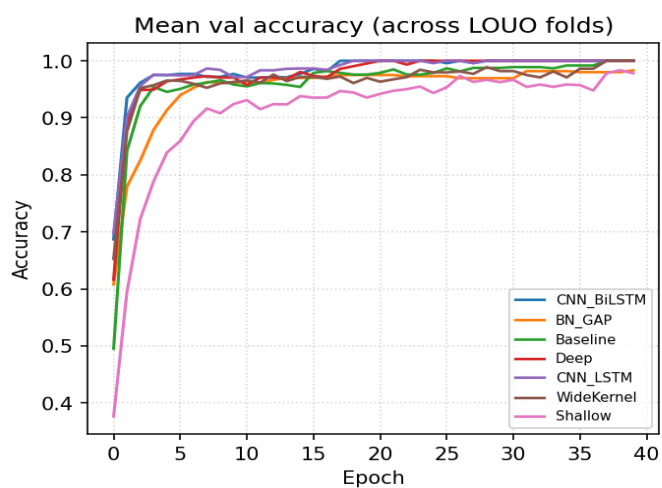
Variant	Params	LOUO mean	LOUO std	Folds	Held-out user acc
CNN_BiLSTM	105,510	0.9821	0.0249	13	—
BN_GAP	57,190	0.9795	0.0463	13	—
Baseline	51,750	0.9718	0.0450	13	—
Deep	72,806	0.9699	0.0386	13	—
CNN_LSTM	68,390	0.9686	0.0581	13	—
WideKernel	70,182	0.9661	0.0321	13	—
Shallow	41,734	0.9635	0.0402	13	—

Best LOUO mean: CNN_BiLSTM

LOUO accuracy per variant

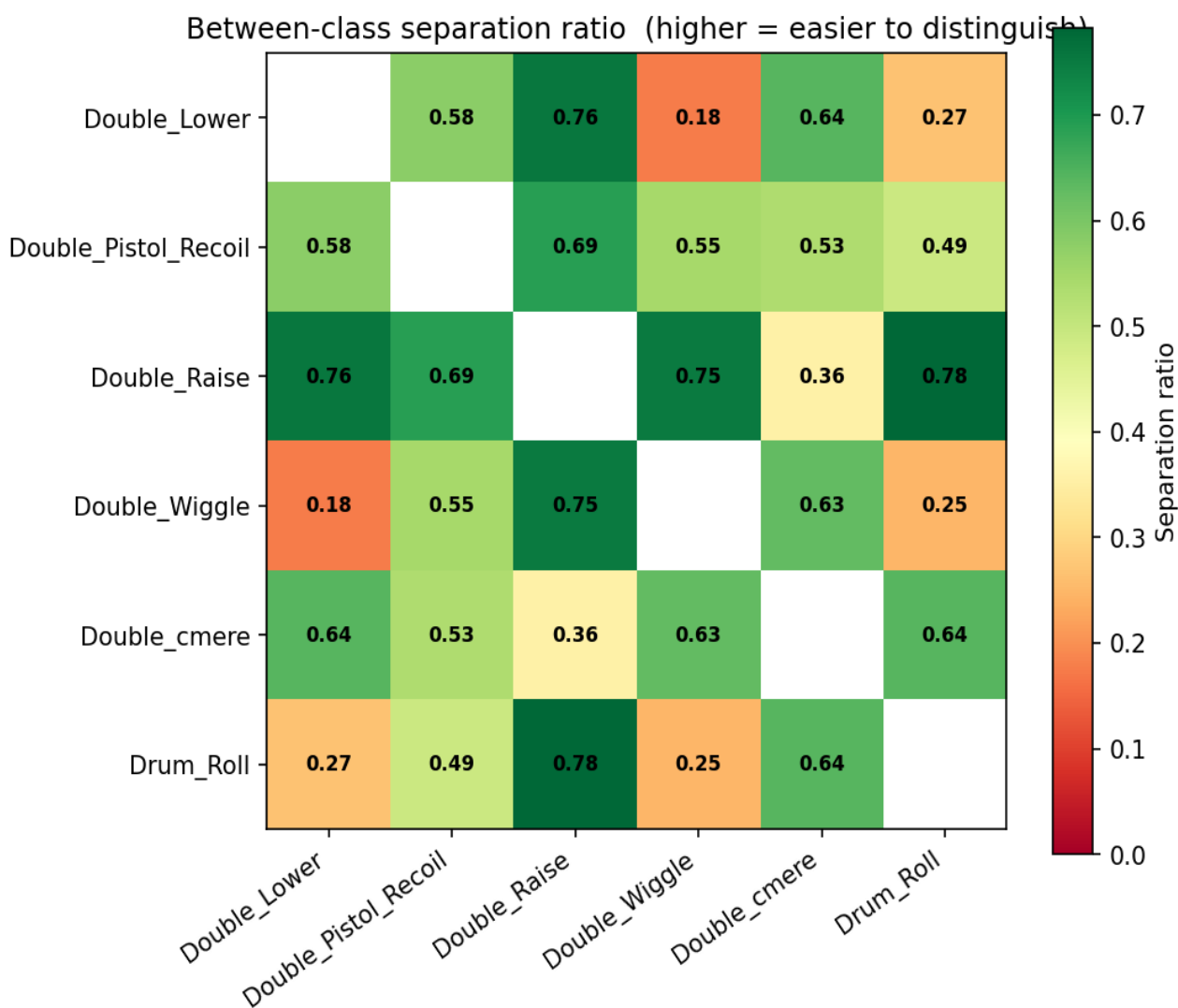
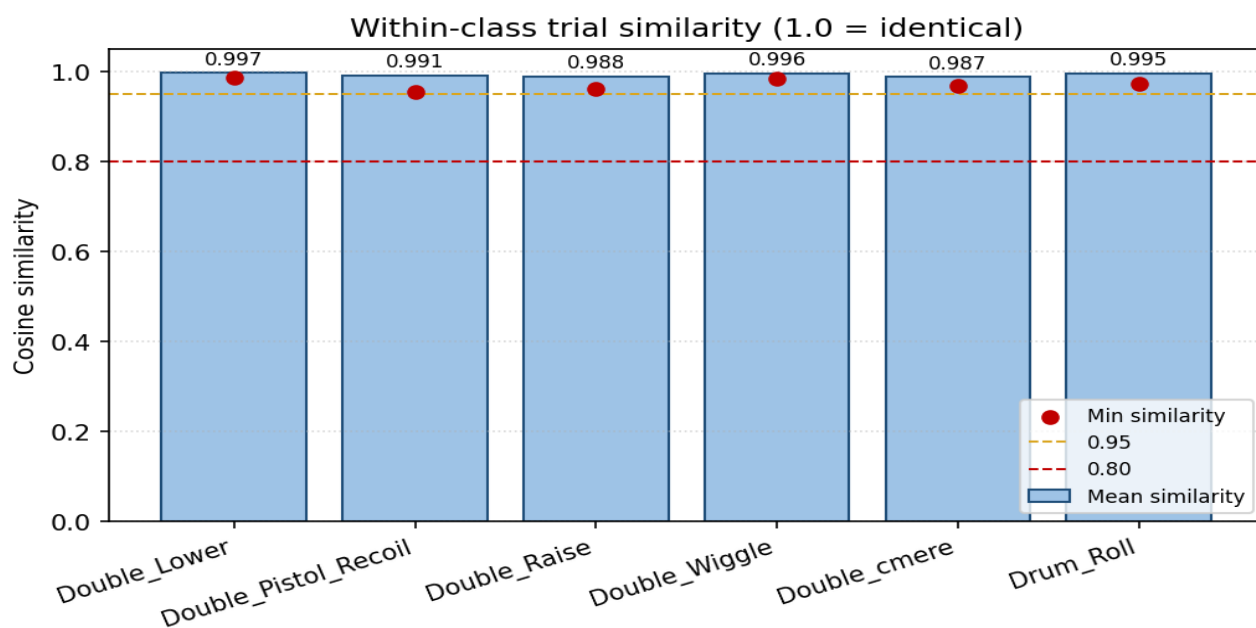


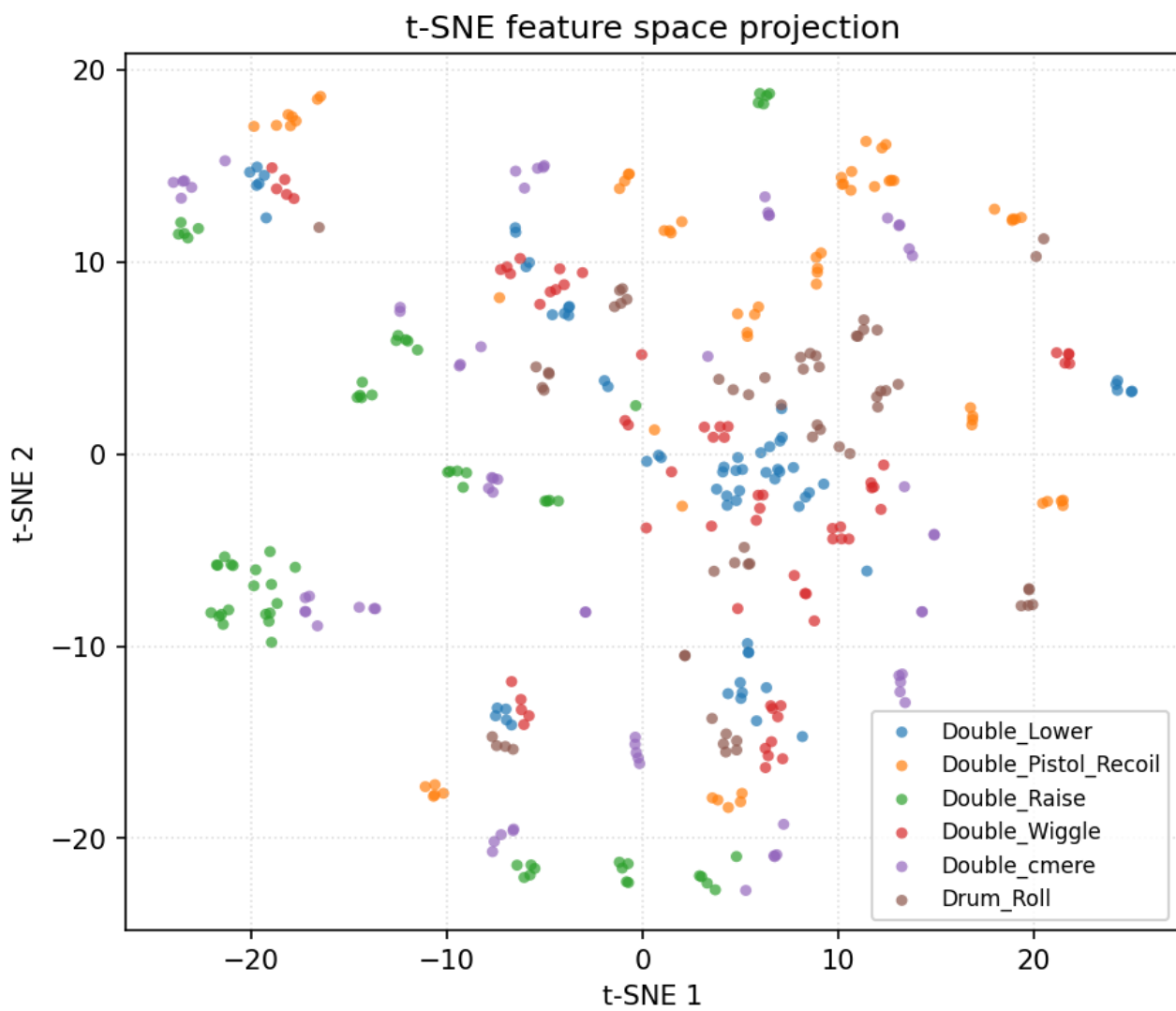
Mean training curves across LOUO folds



Data diagnostics — pooled across users

Filtered + resampled trials from every LOUO user (before per-trial normalisation).

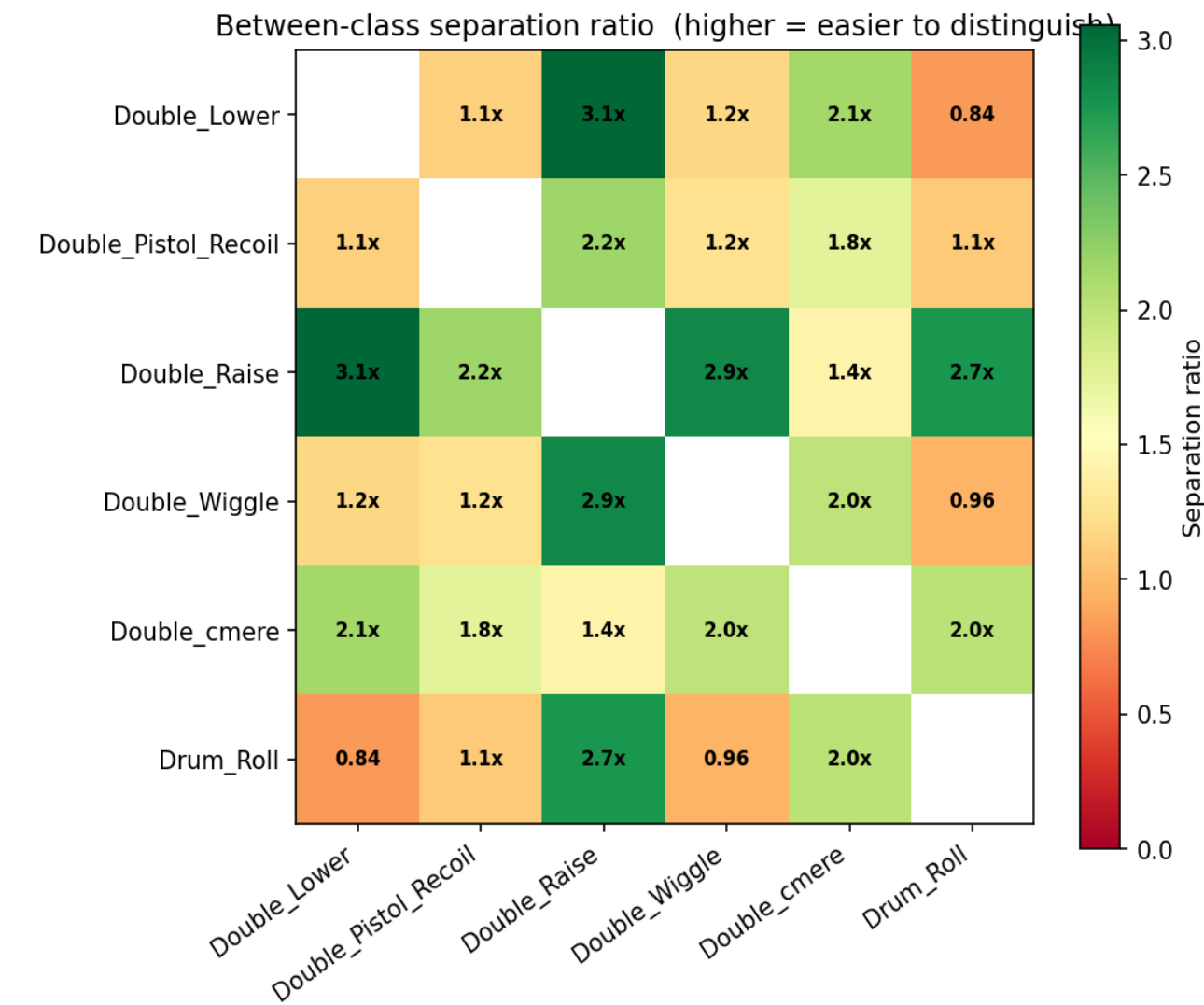
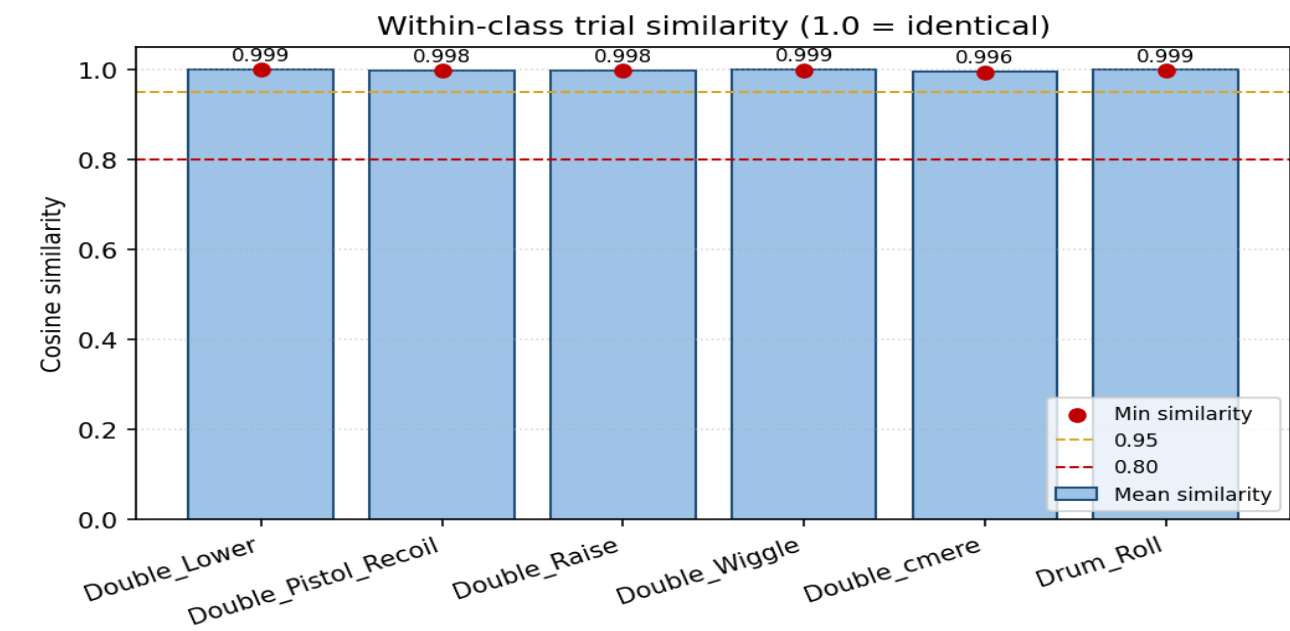




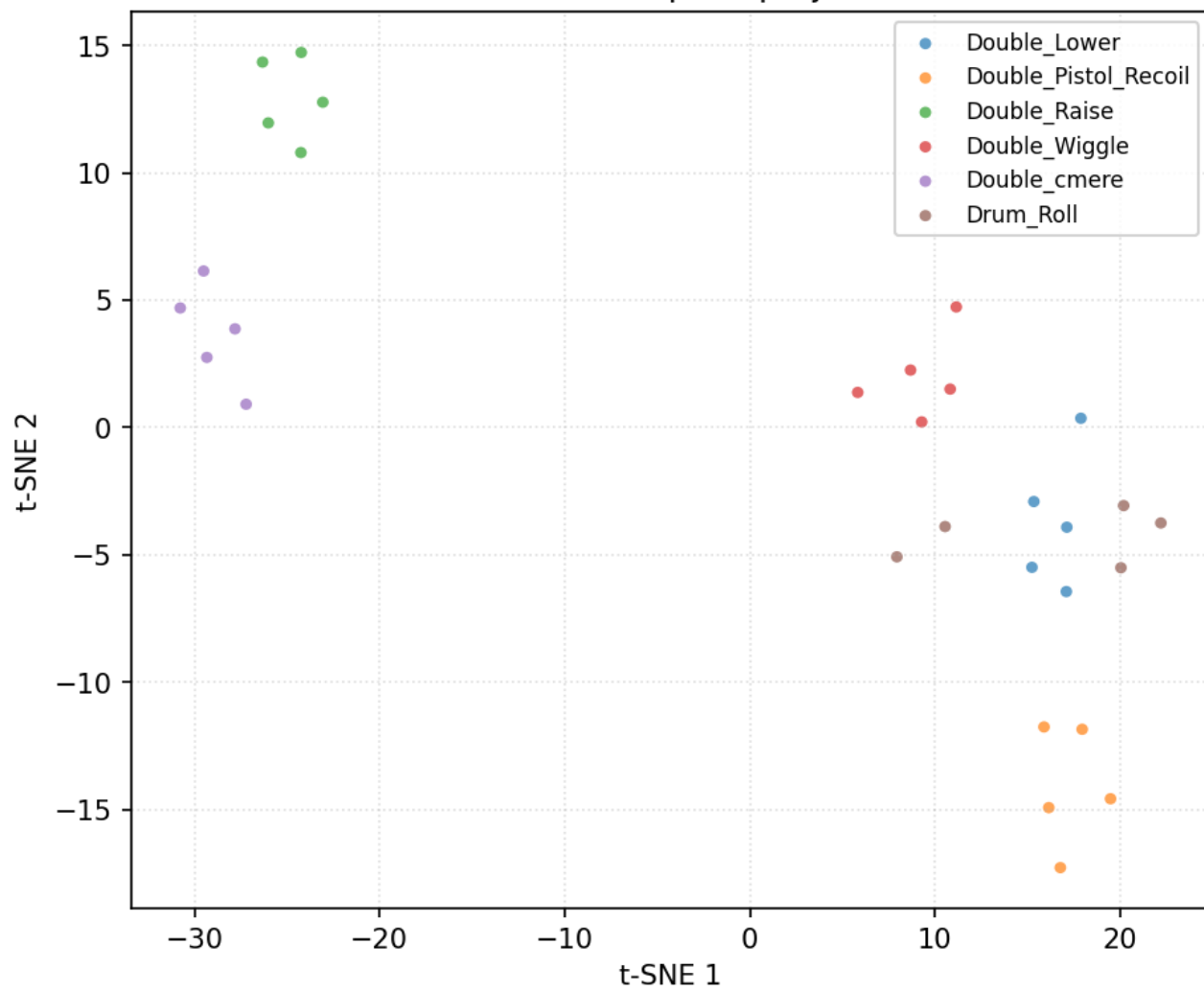
Reading guide: within-class similarity > 0.95 = very consistent; < 0.80 = high variance. Separation ratio > 100x = trivially separable, 20-100x = easy, 5-20x = moderate, < 5x = hard (expect confusion).

Data diagnostics — user 11_Mansh

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

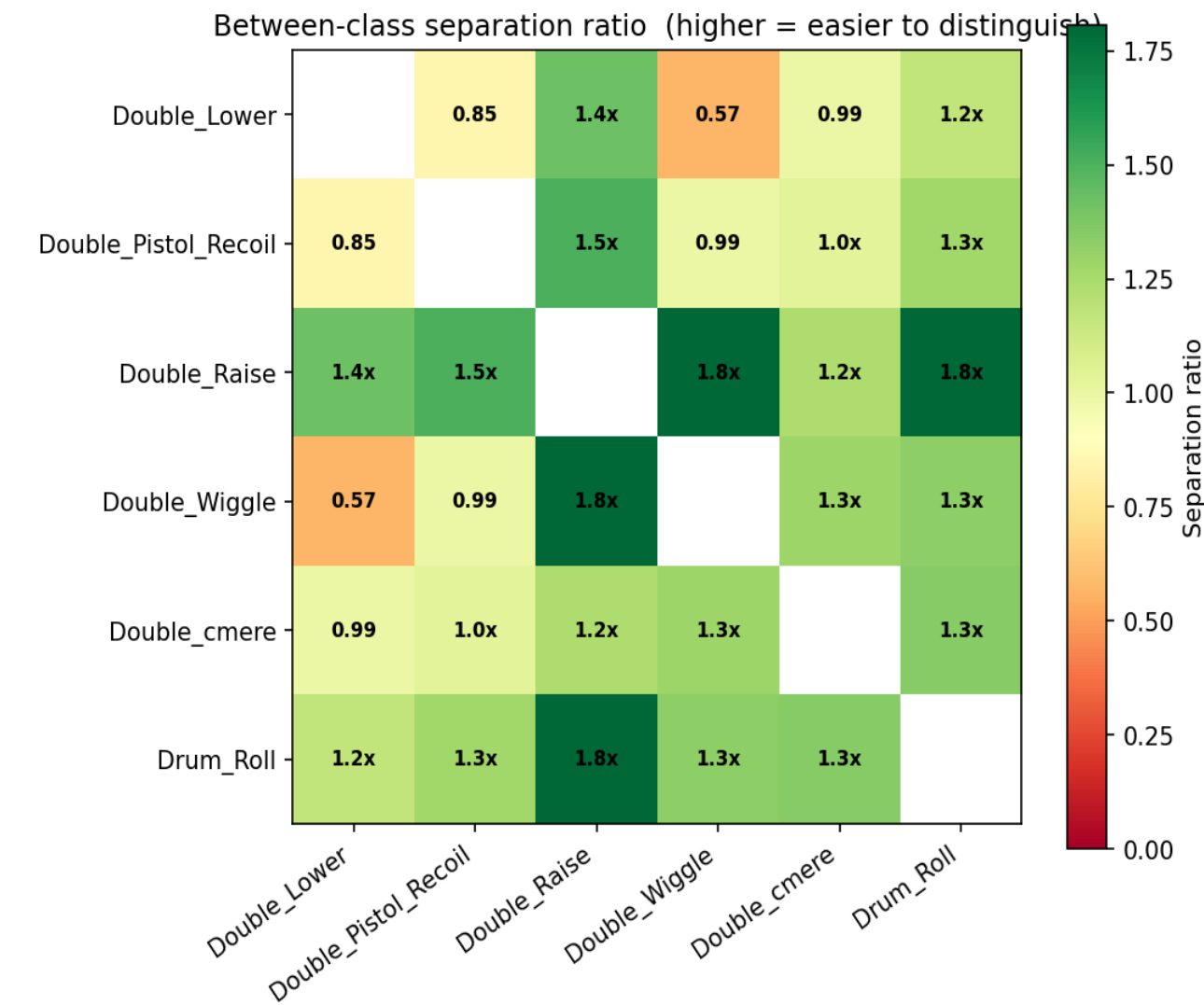
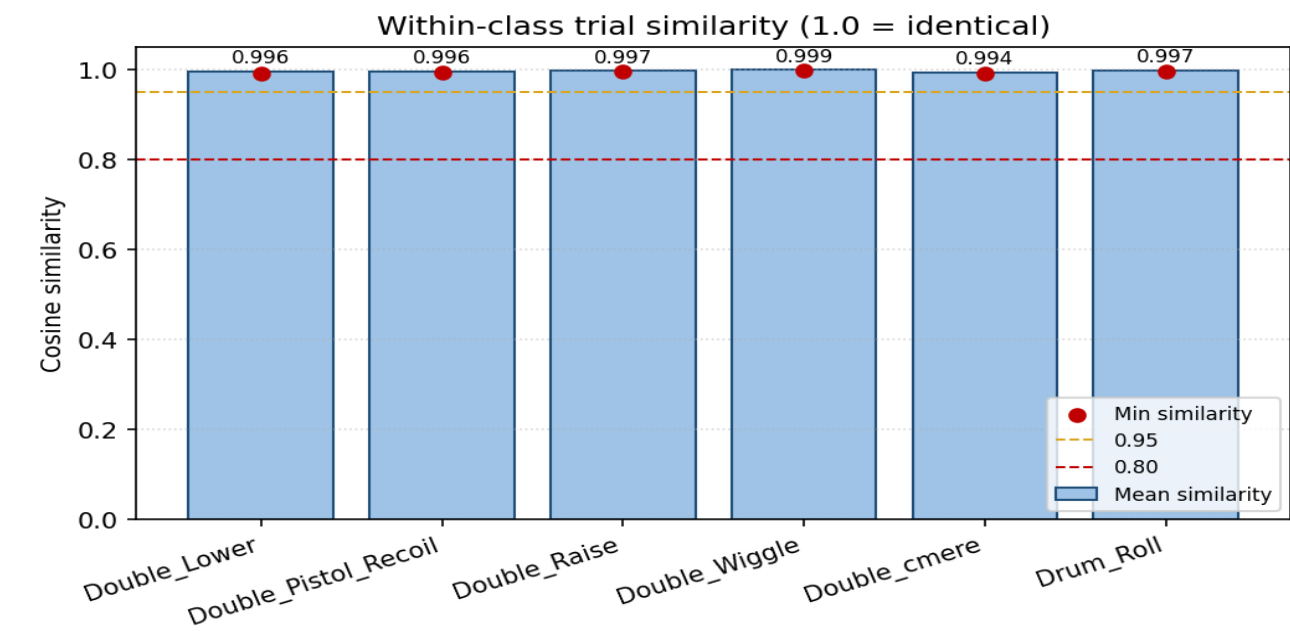


t-SNE feature space projection

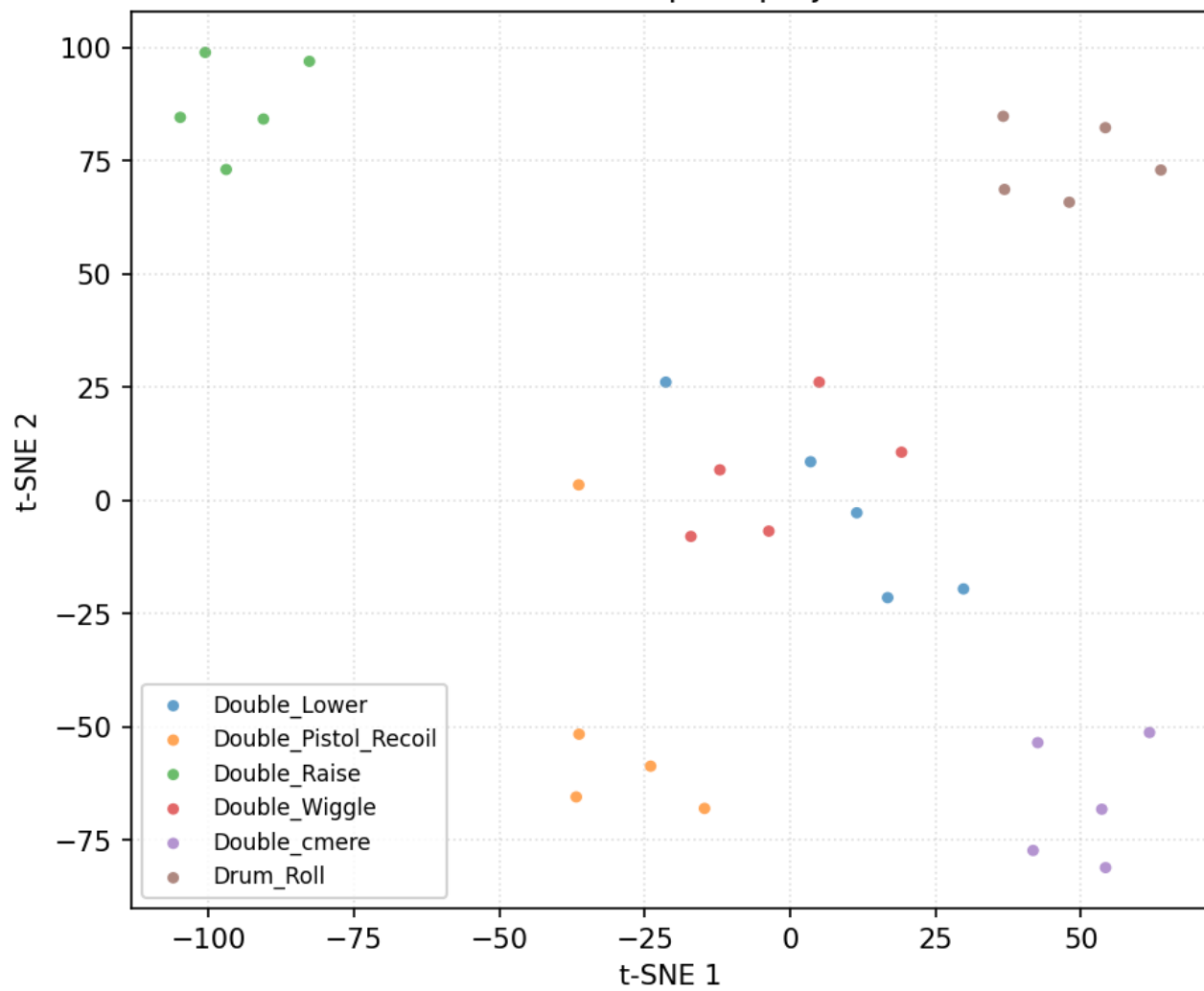


Data diagnostics — user 12_Marcus

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

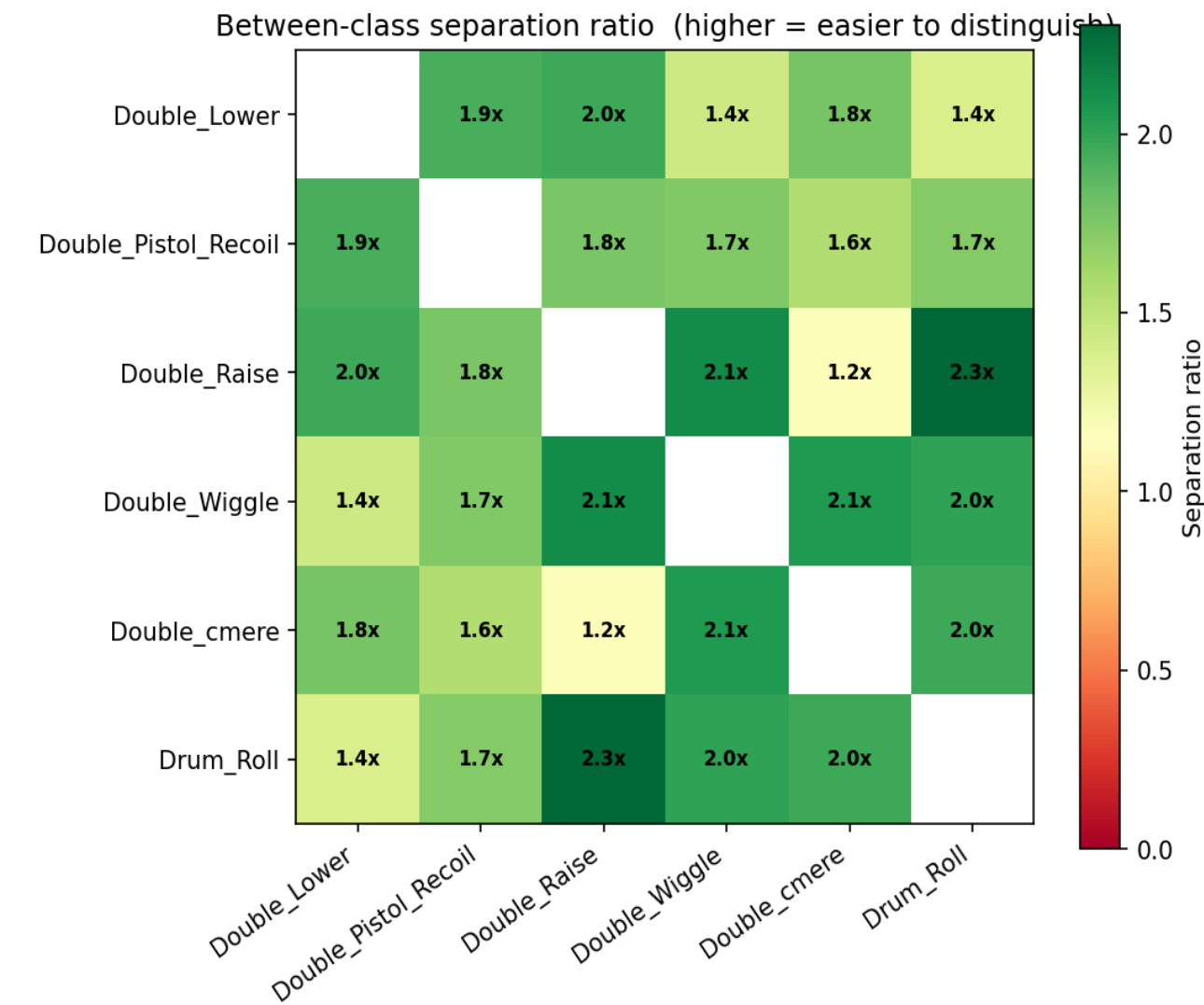
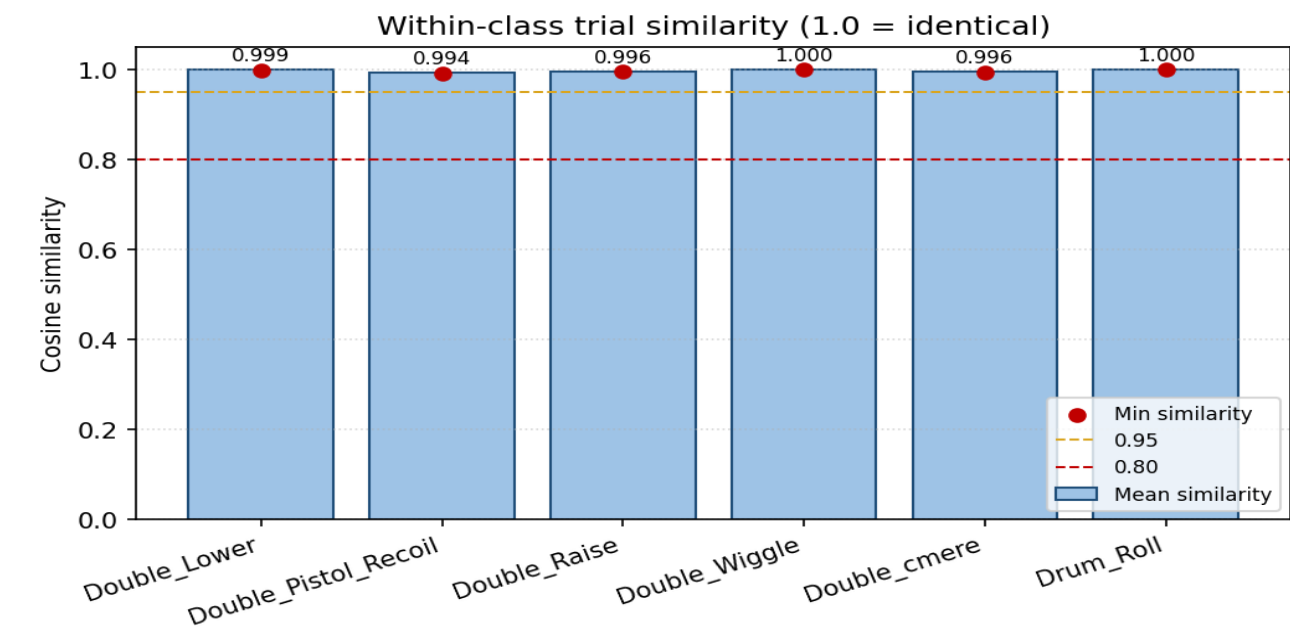


t-SNE feature space projection

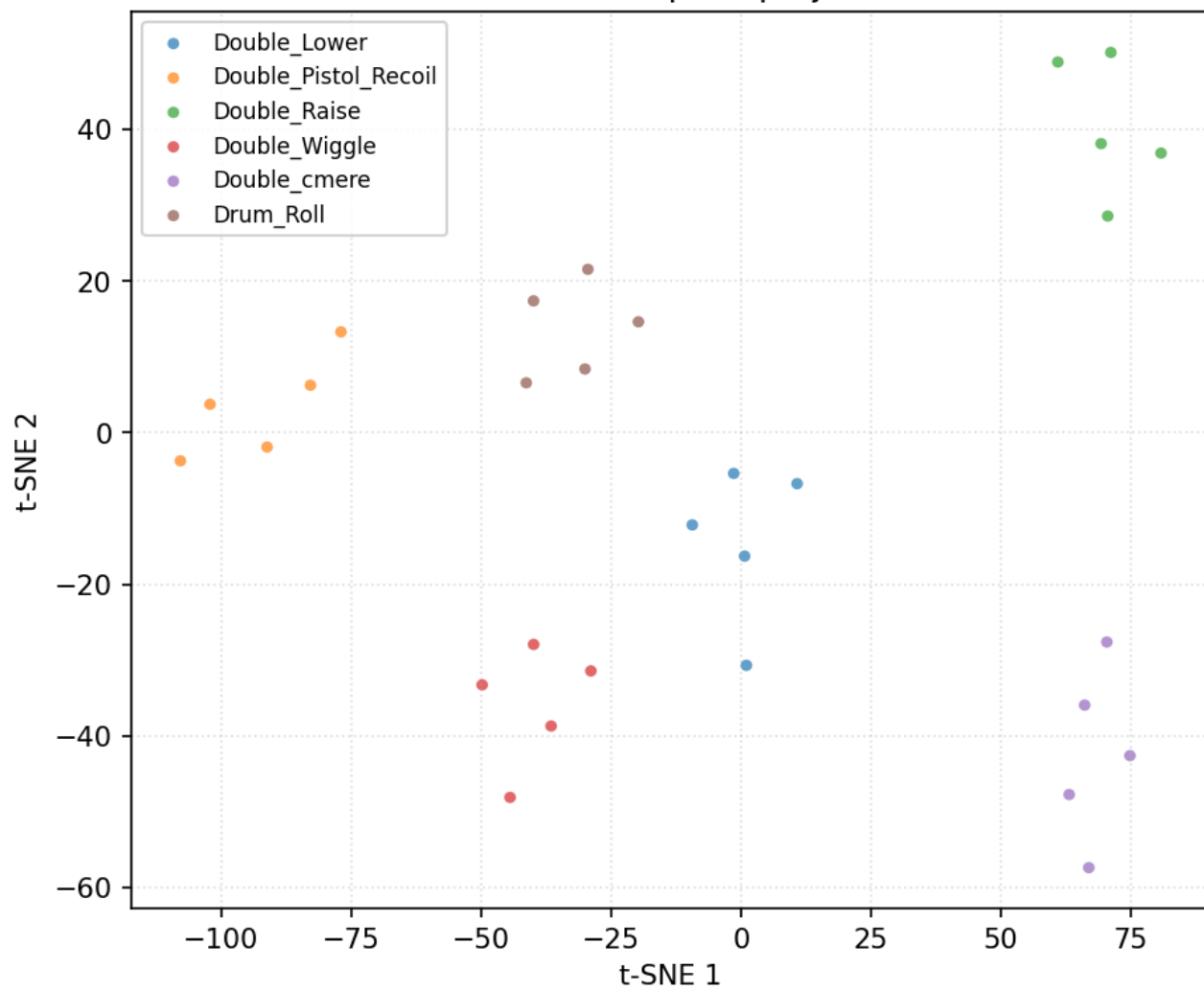


Data diagnostics — user 14_StephenV2

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

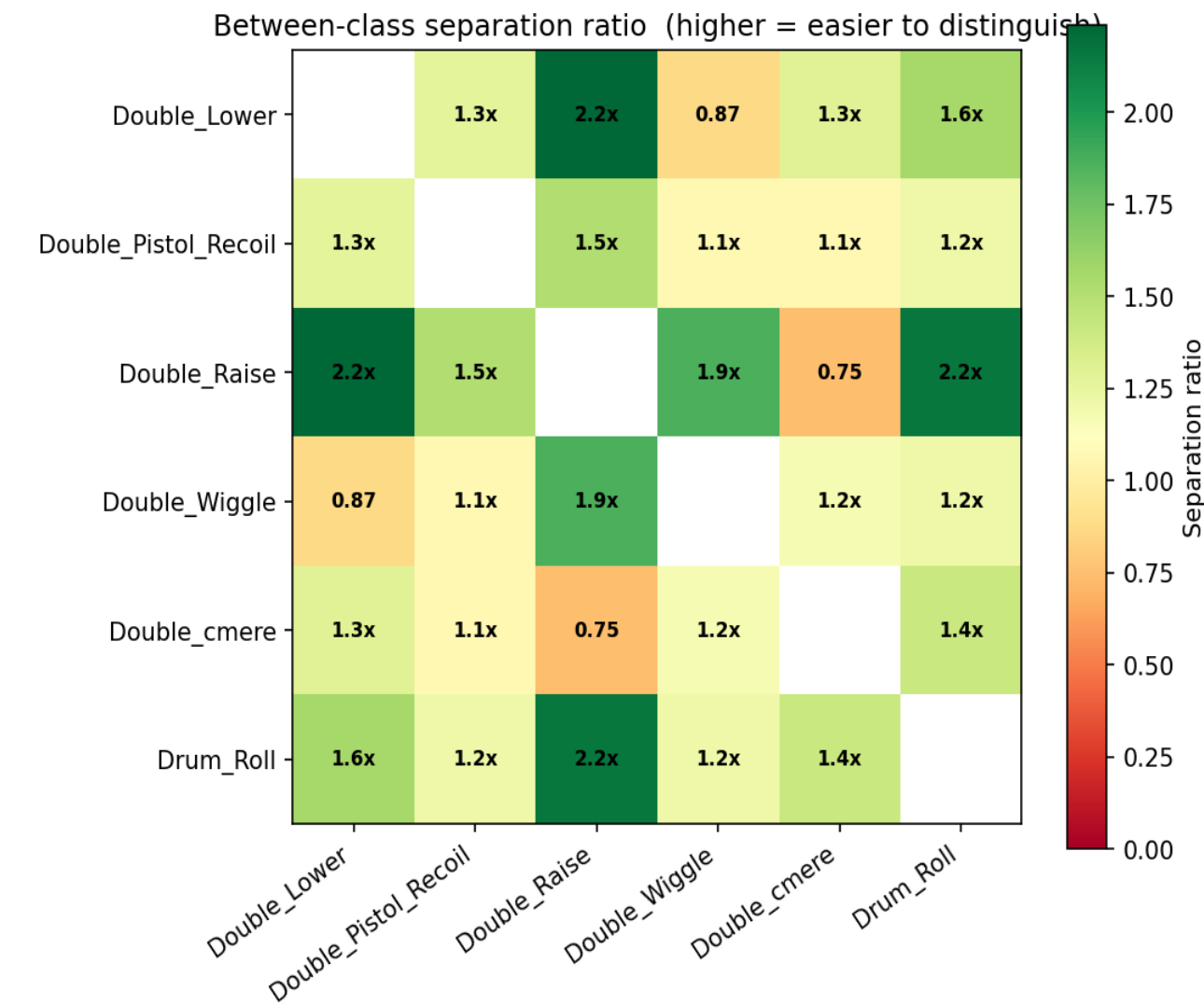
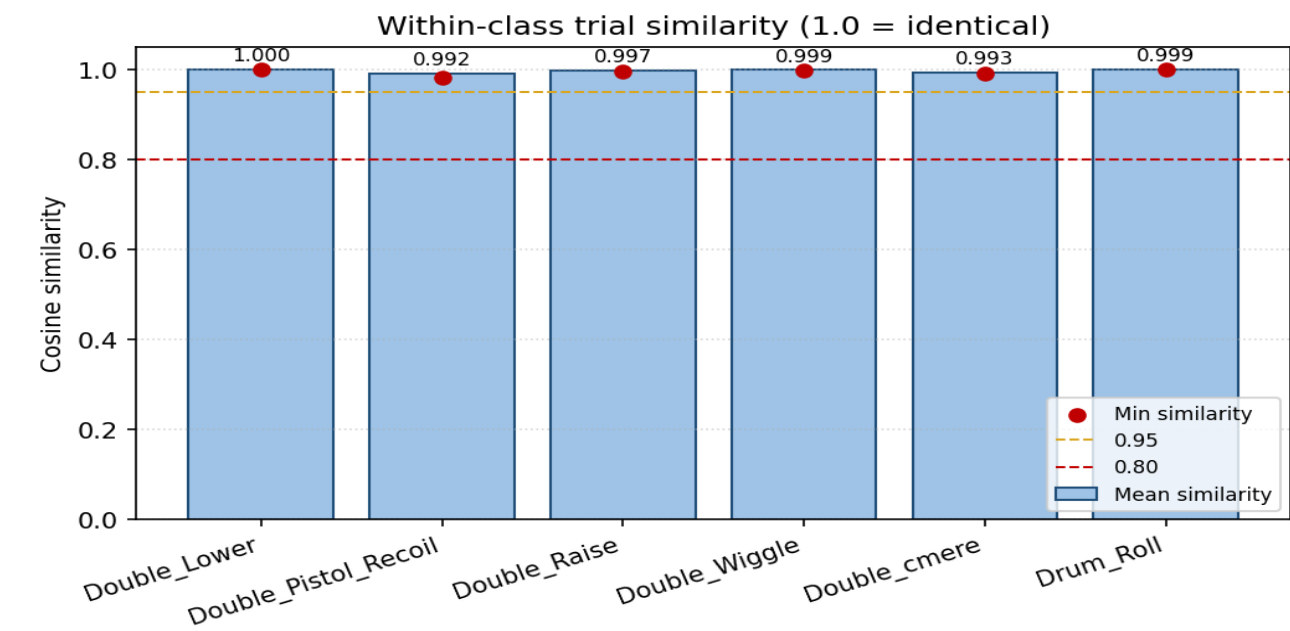


t-SNE feature space projection

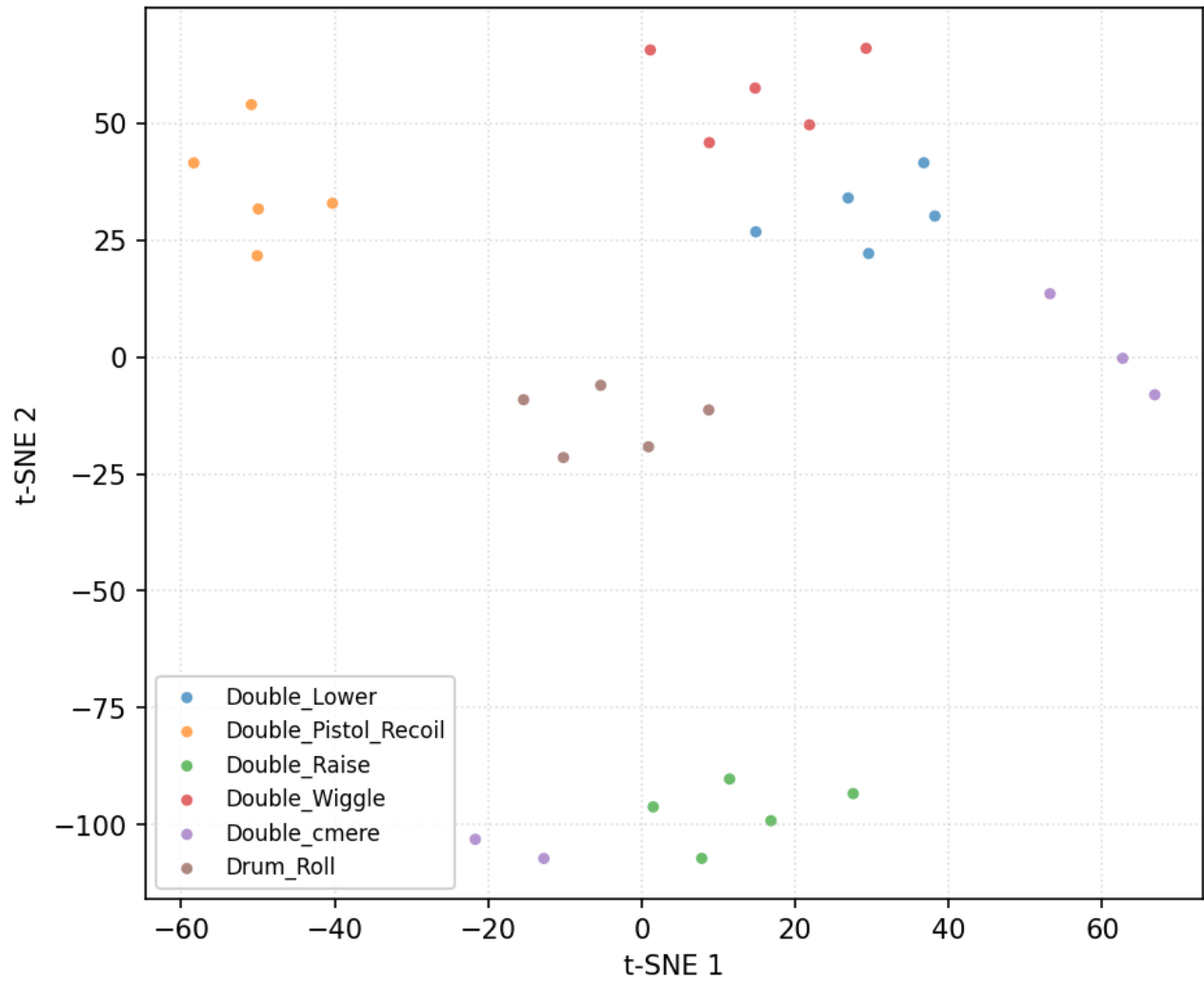


Data diagnostics — user 15_Tash

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

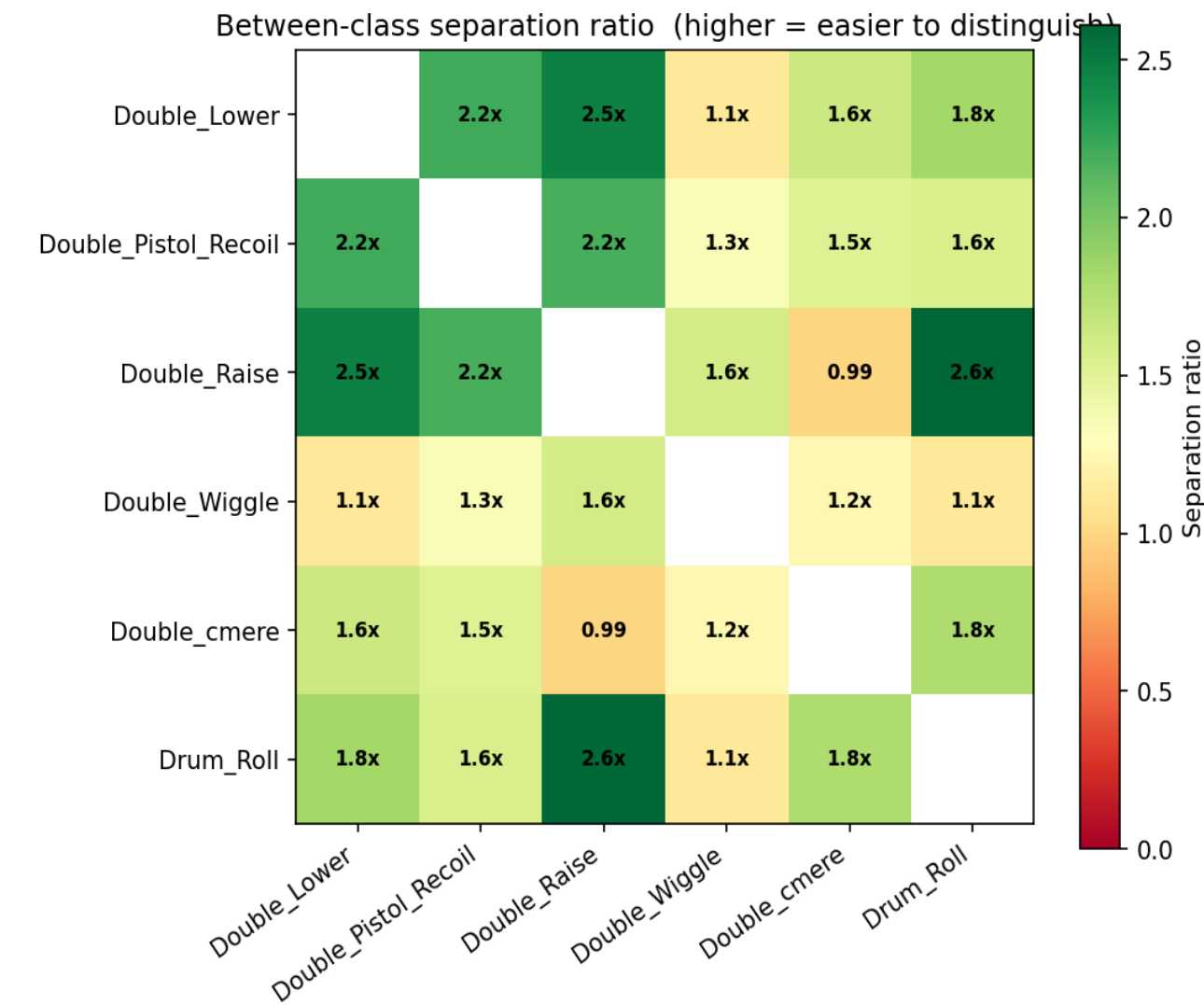
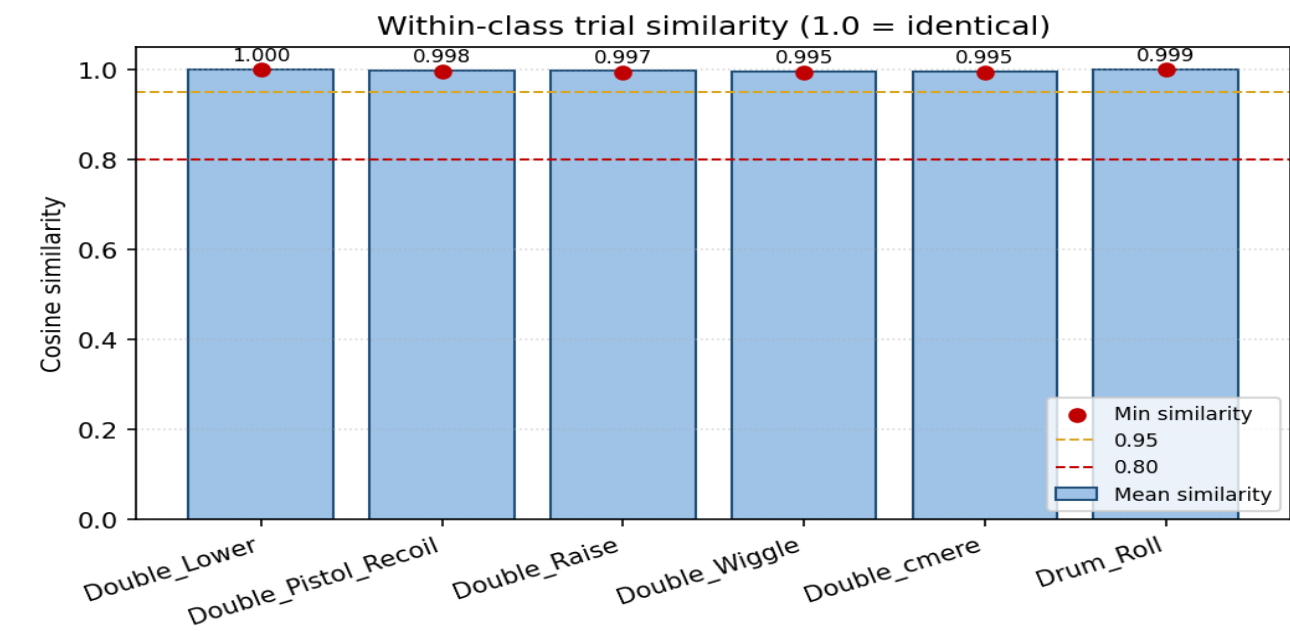


t-SNE feature space projection

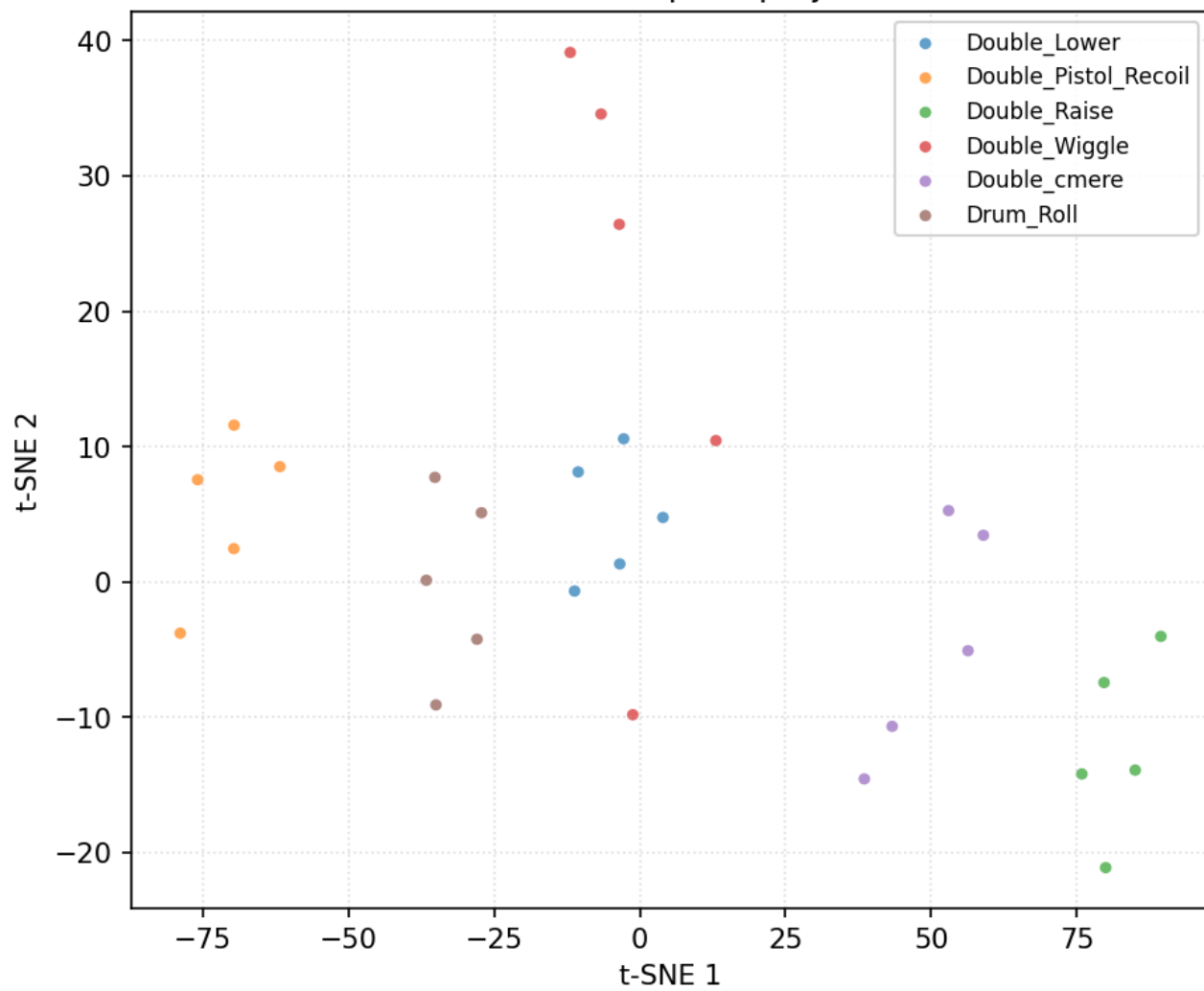


Data diagnostics — user 16_Bella

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

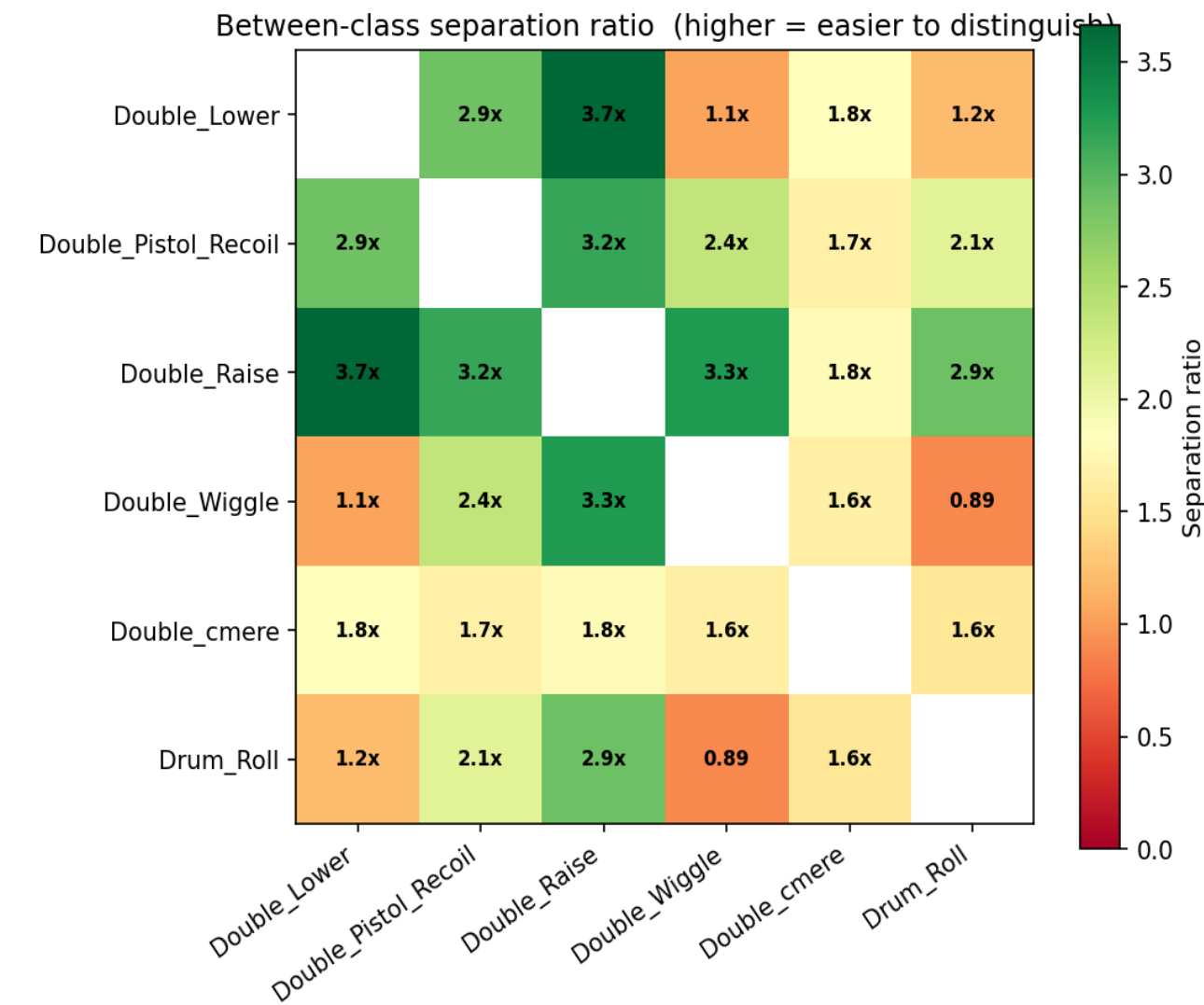
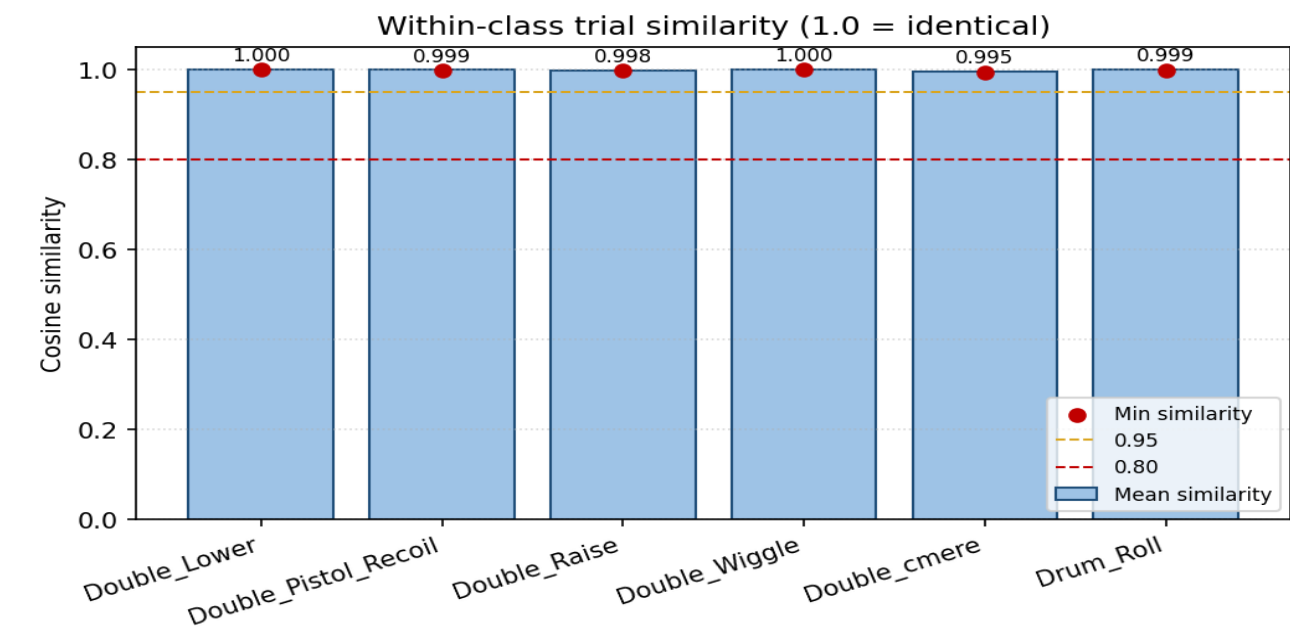


t-SNE feature space projection

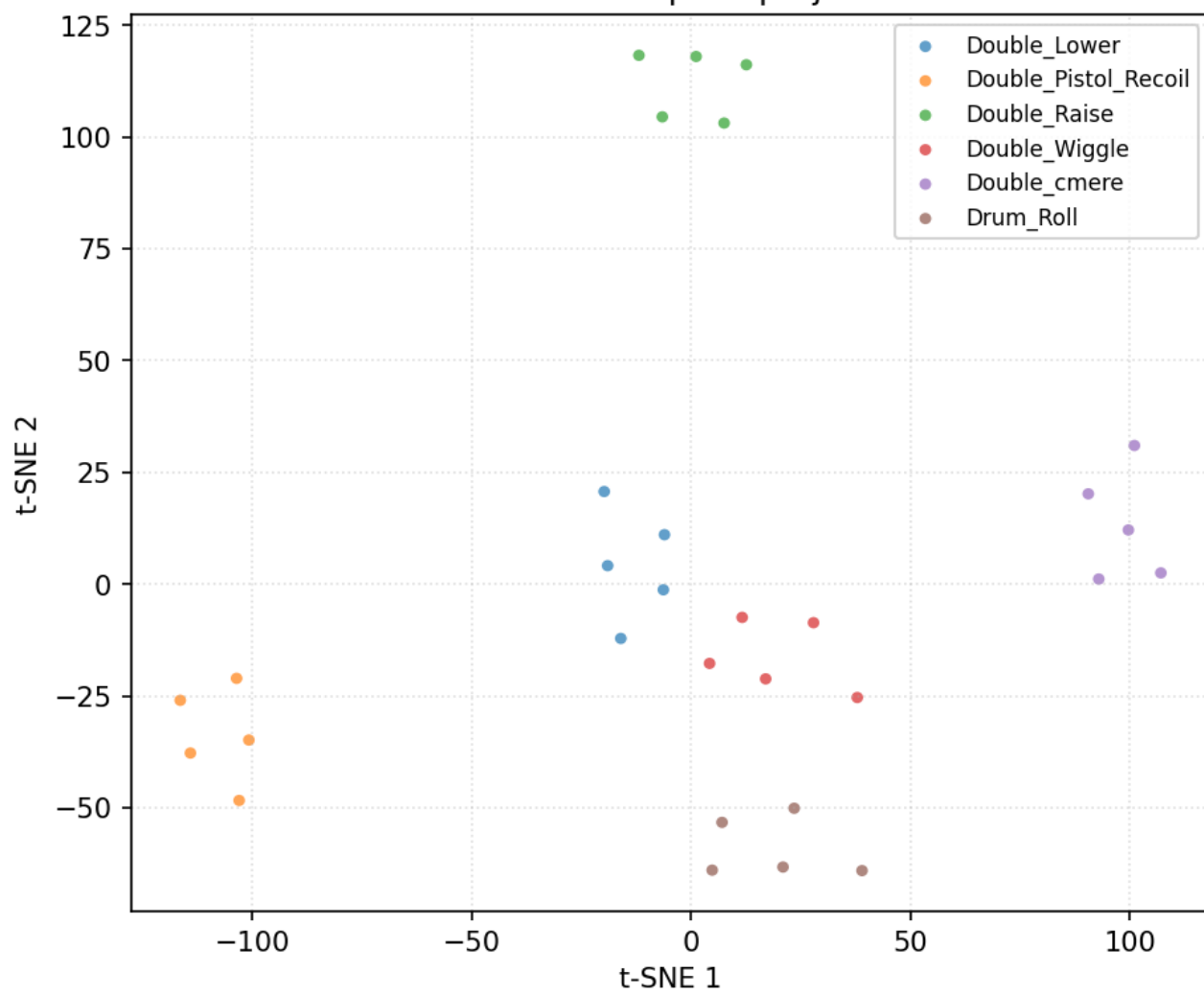


Data diagnostics — user 17_Raquel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

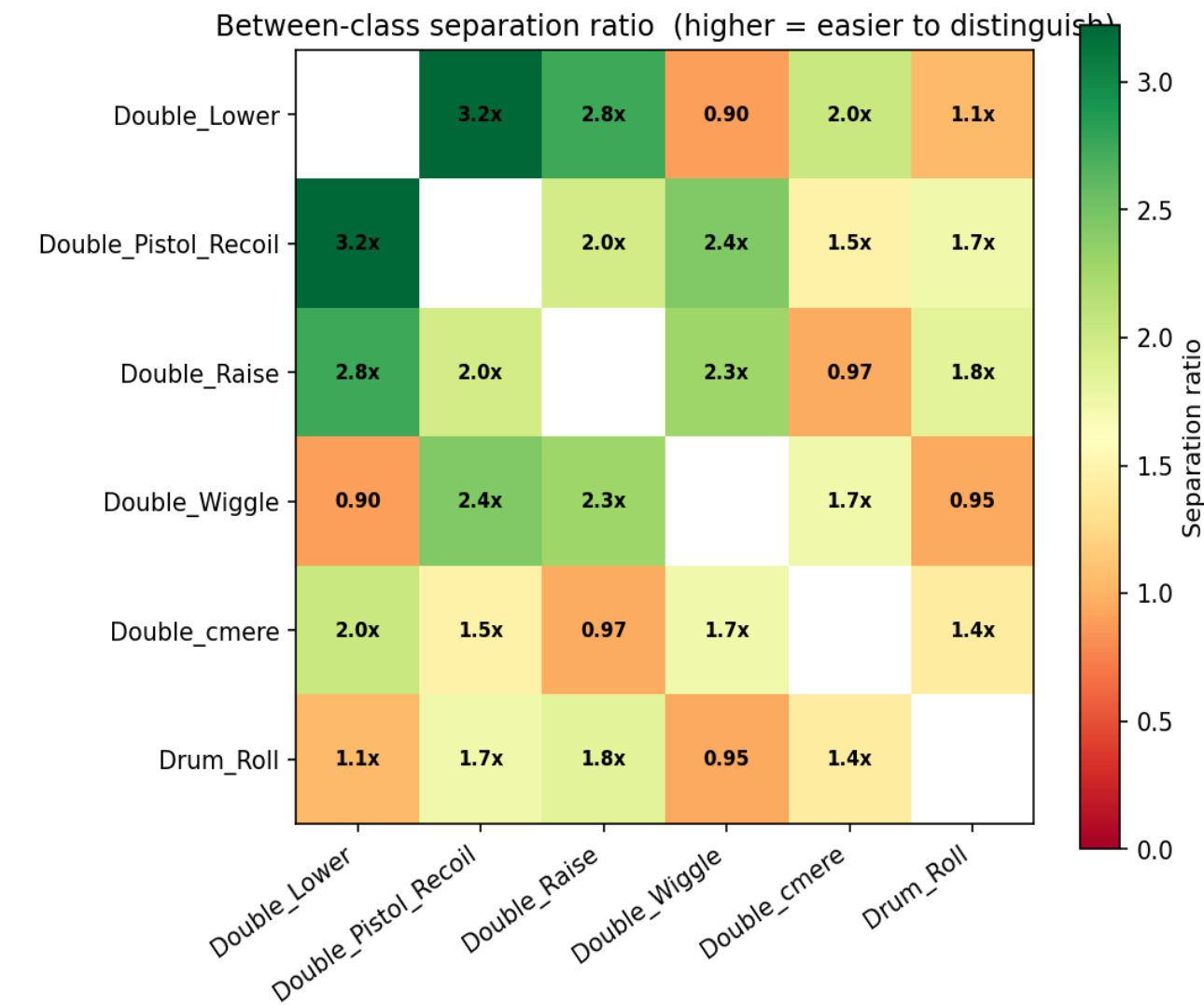
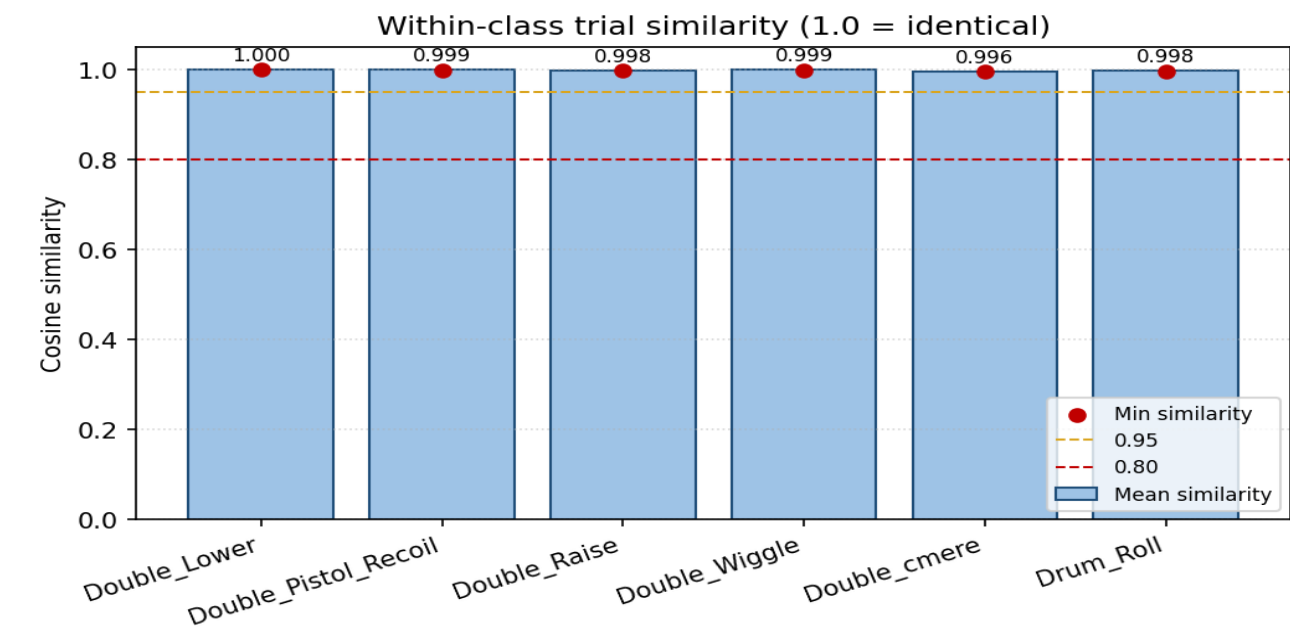


t-SNE feature space projection

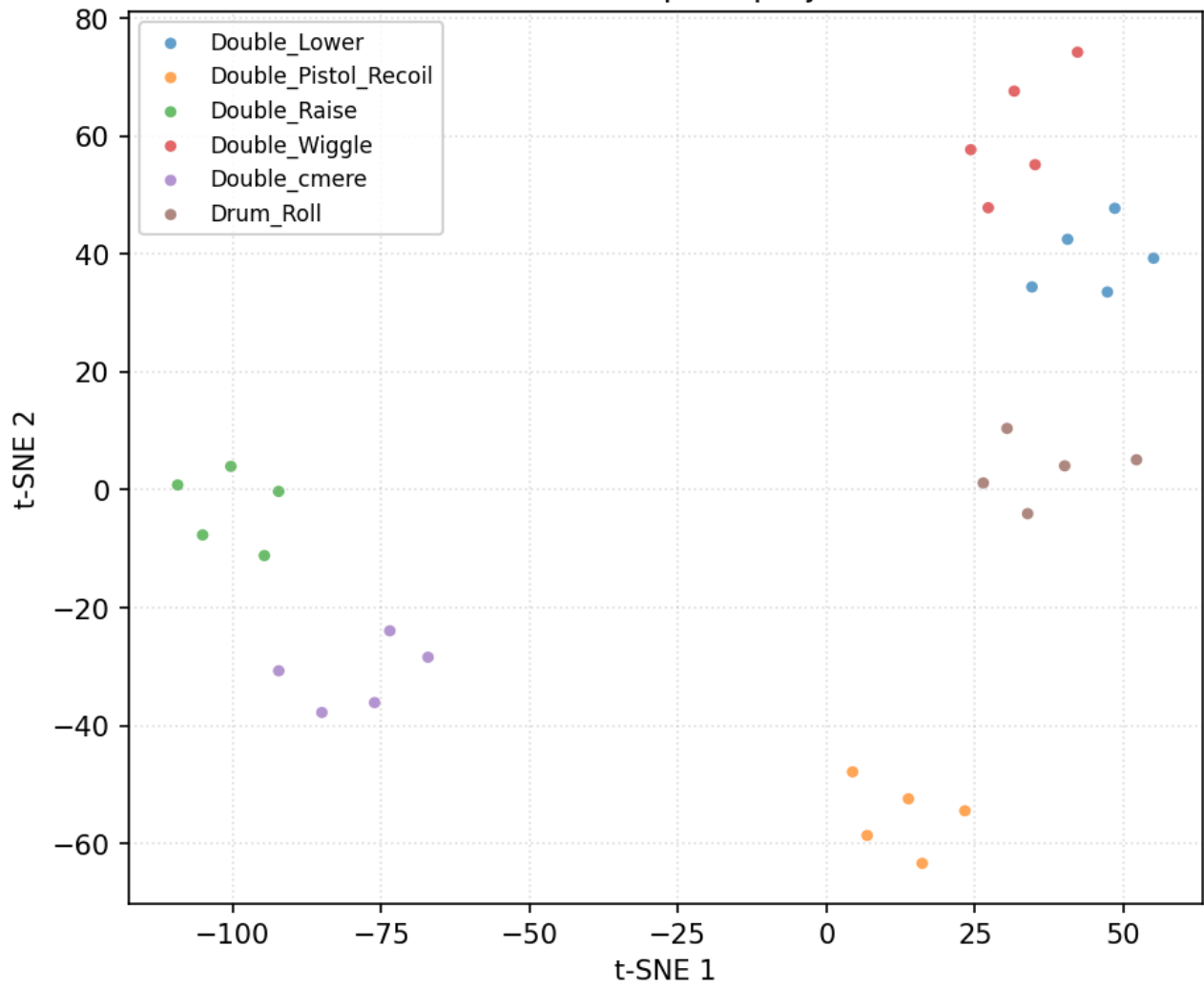


Data diagnostics — user 18_Bridgette

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

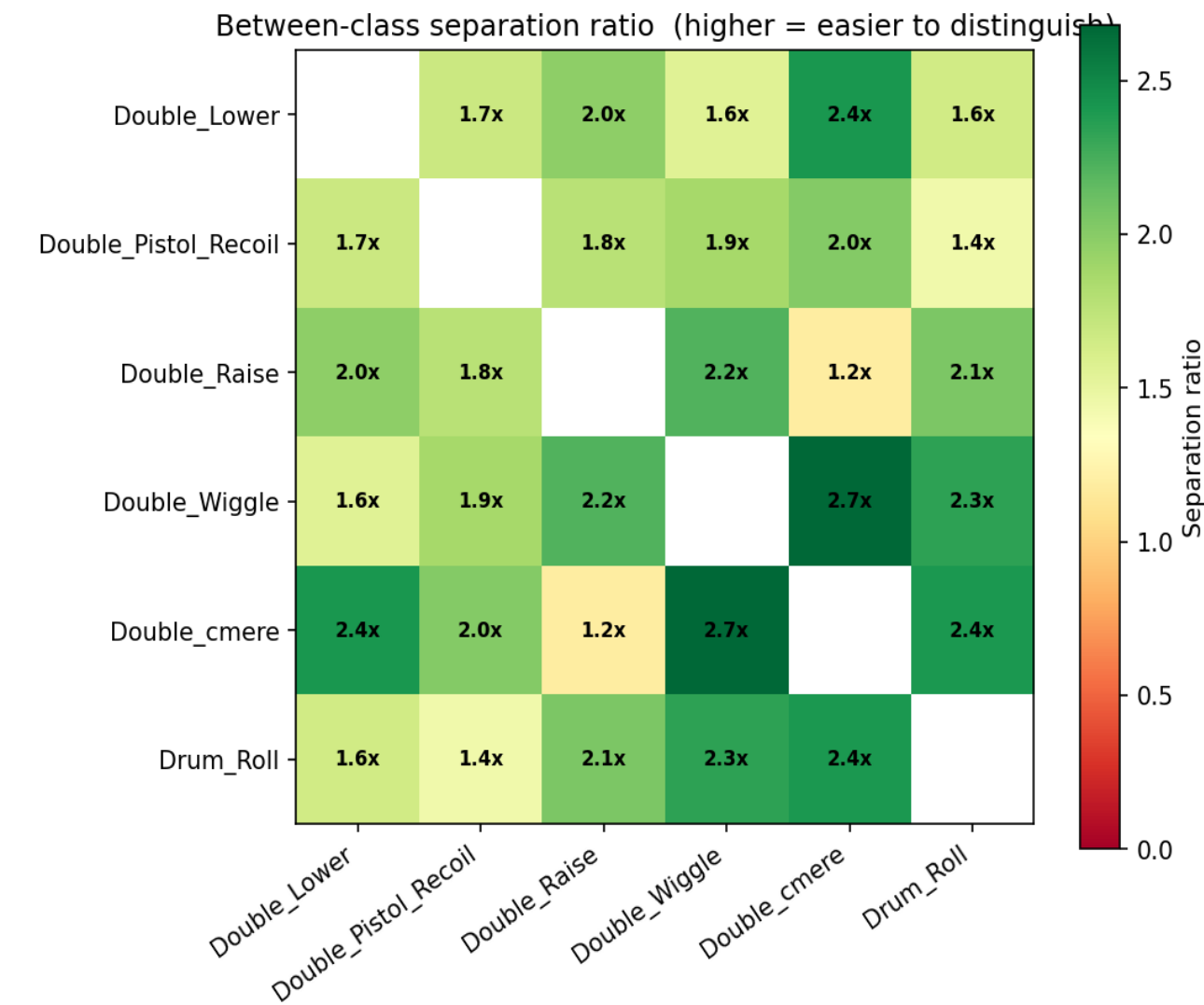
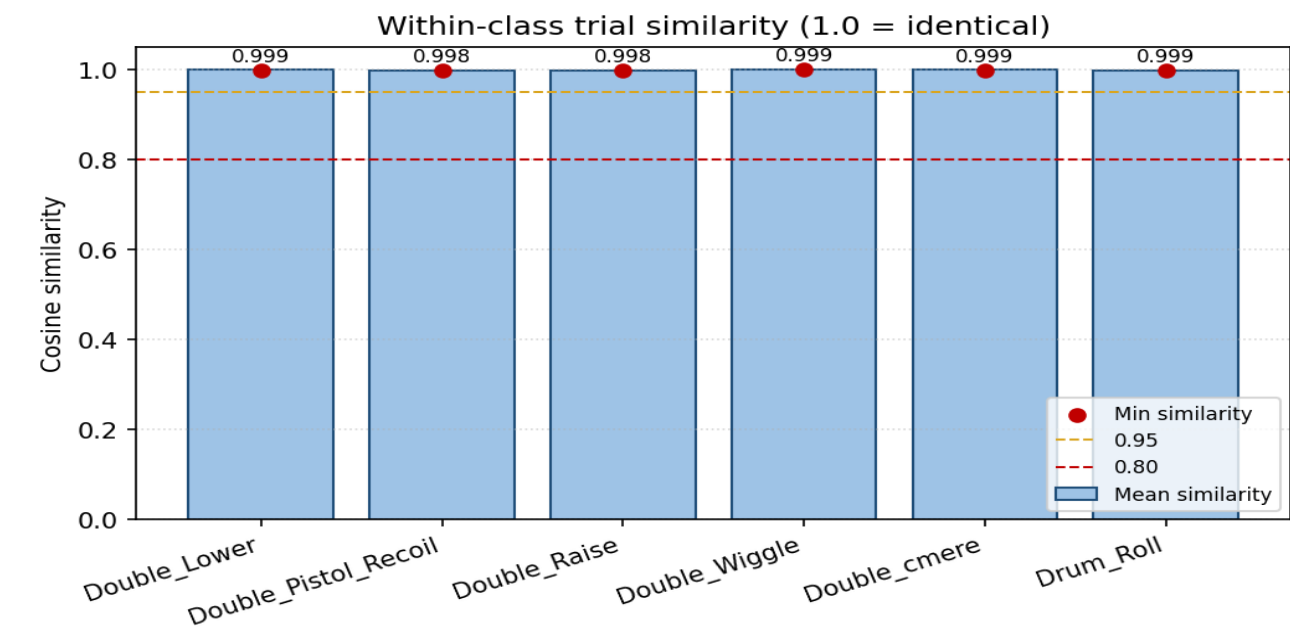


t-SNE feature space projection

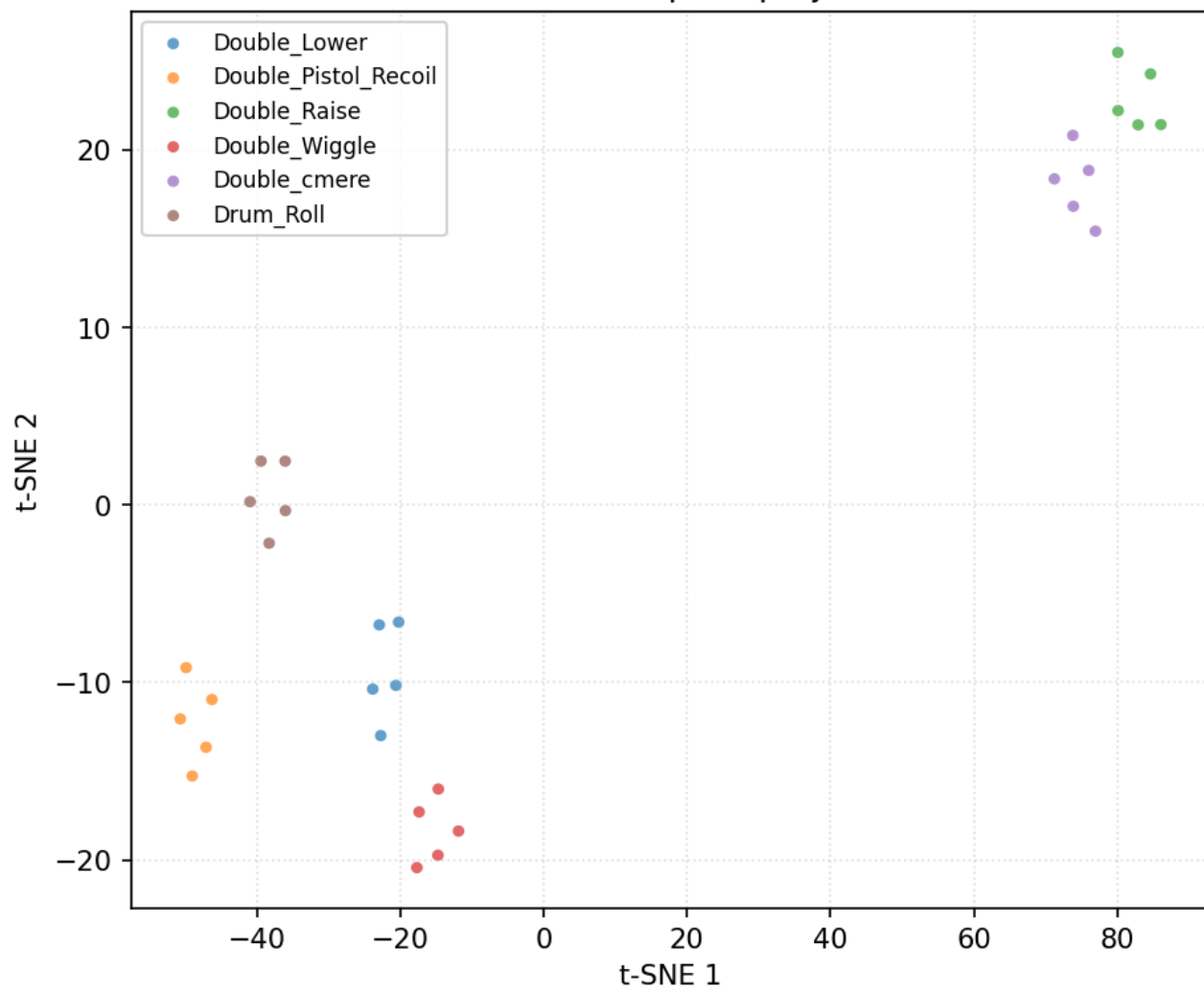


Data diagnostics — user 1_Alán

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

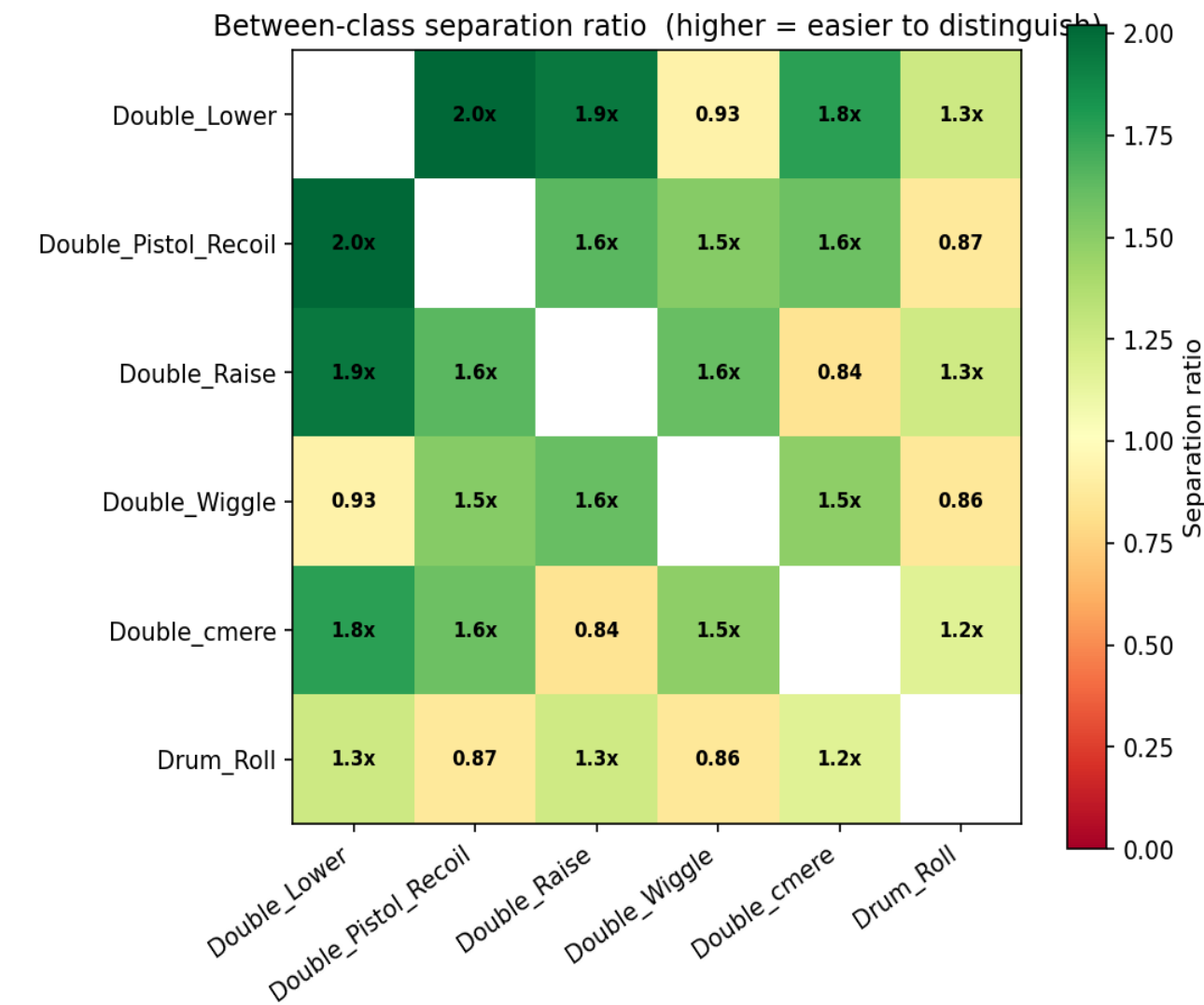
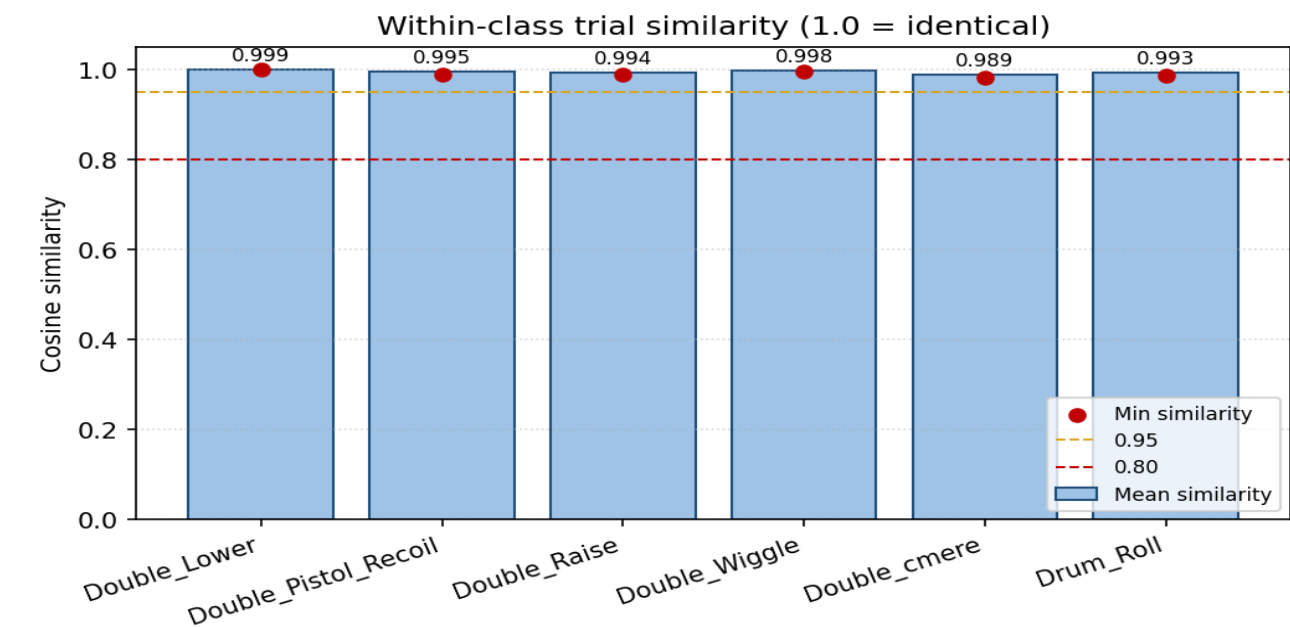


t-SNE feature space projection

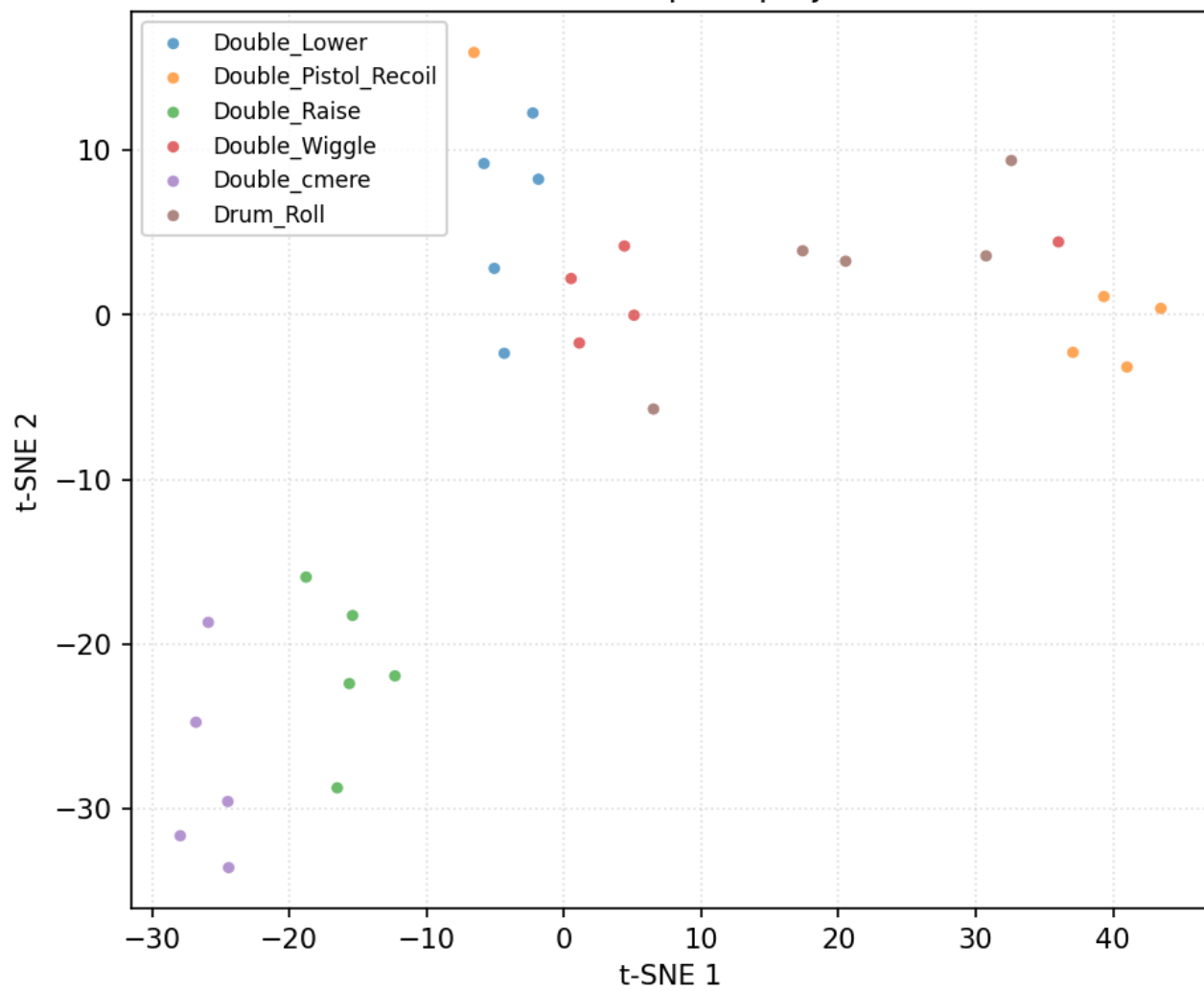


Data diagnostics — user 2_Alex

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

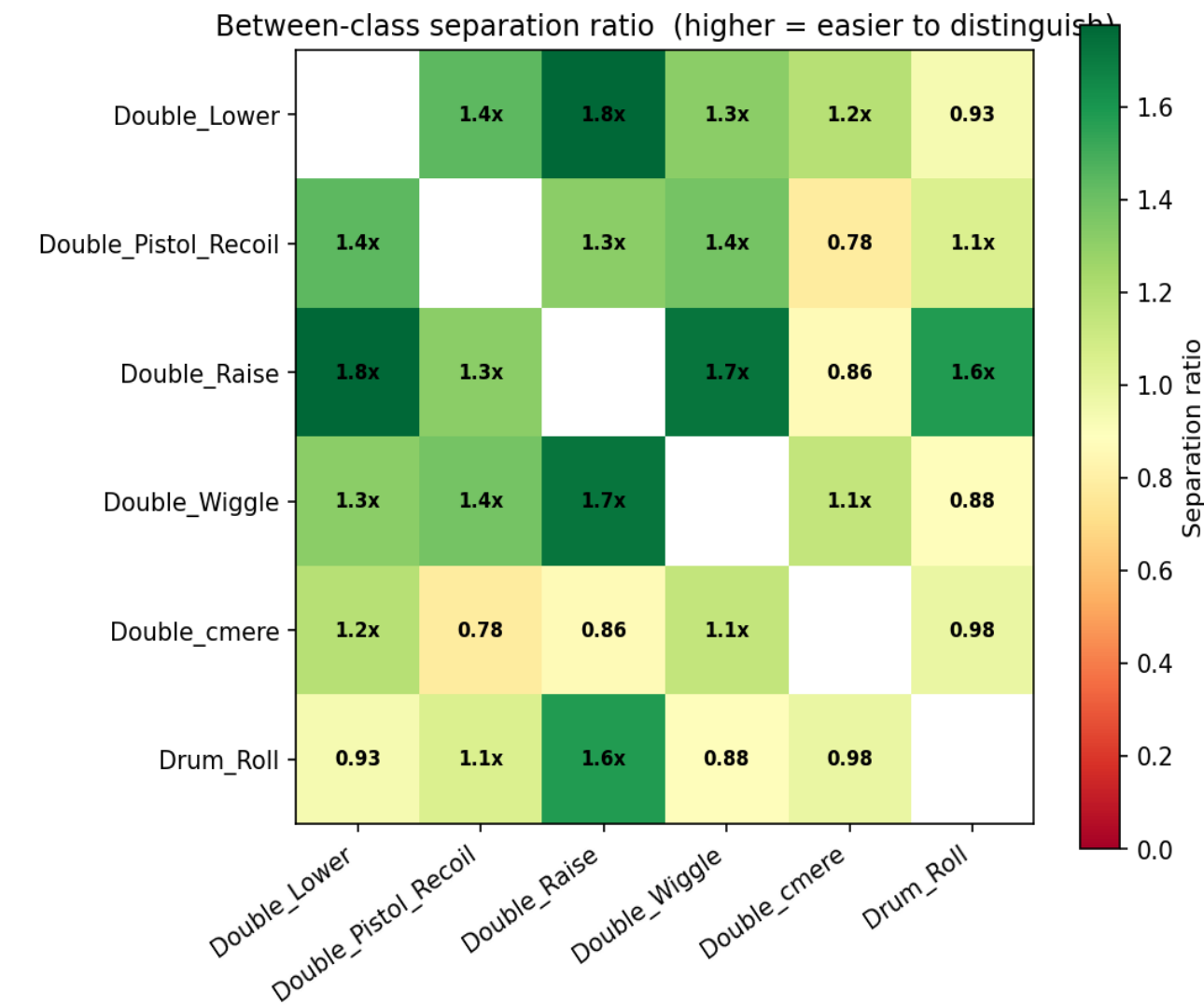
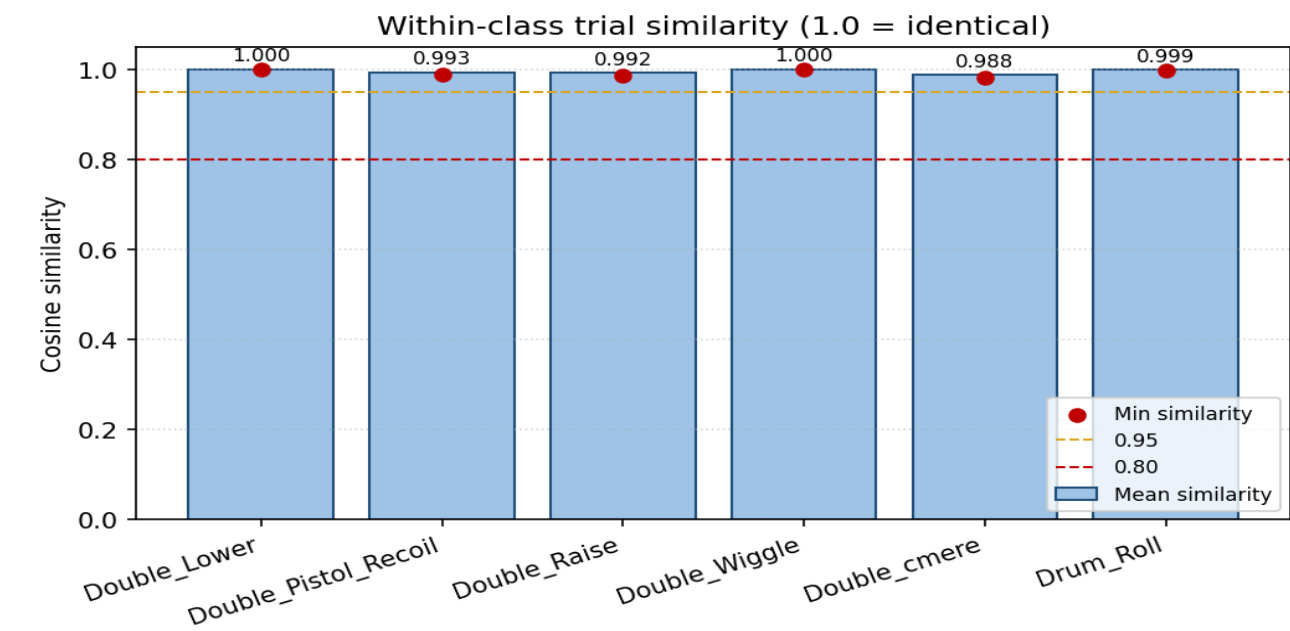


t-SNE feature space projection

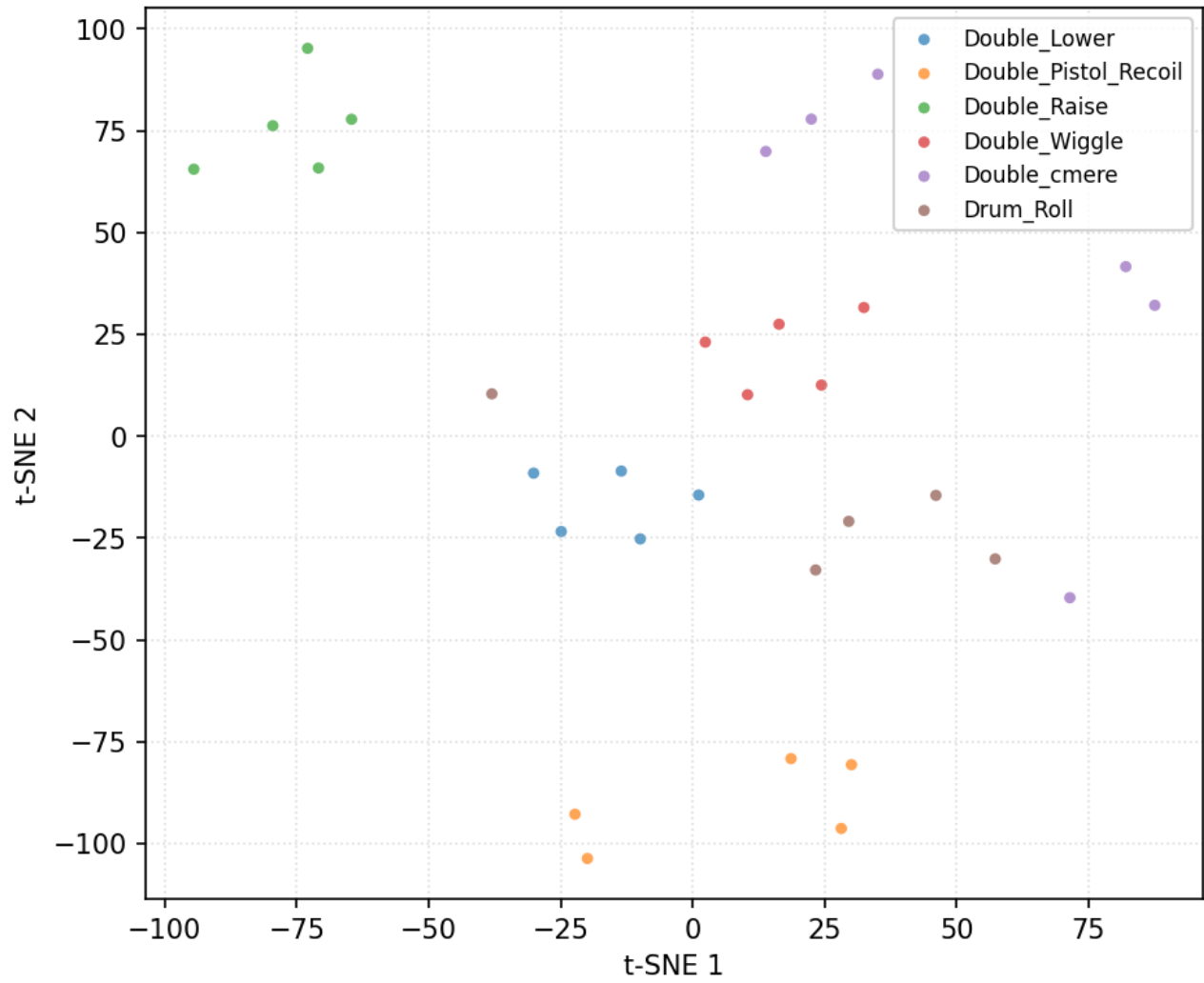


Data diagnostics — user 3_Anghad

31 trials, 6 classes (filtered + resampled, before per-trial normalisation).

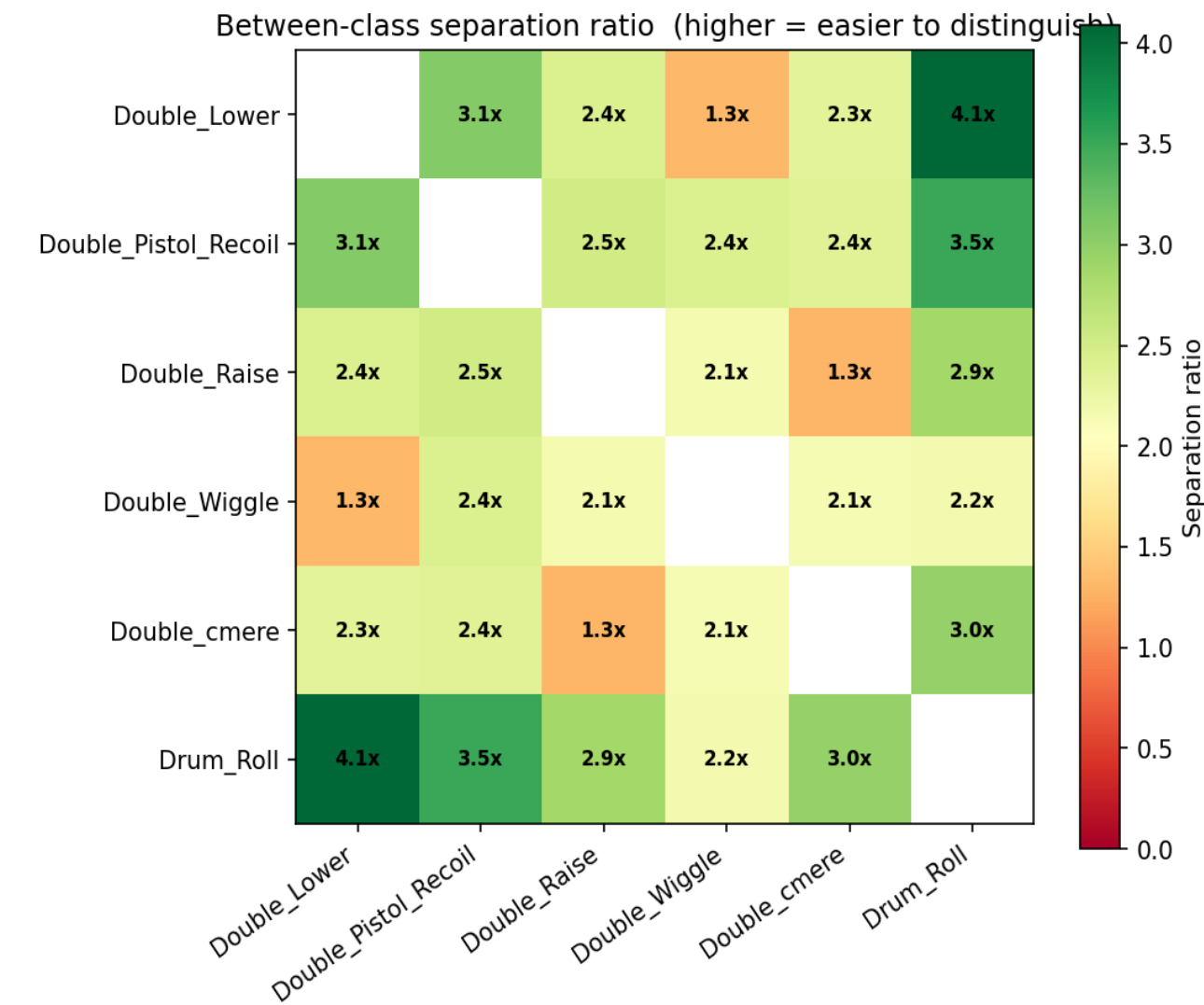
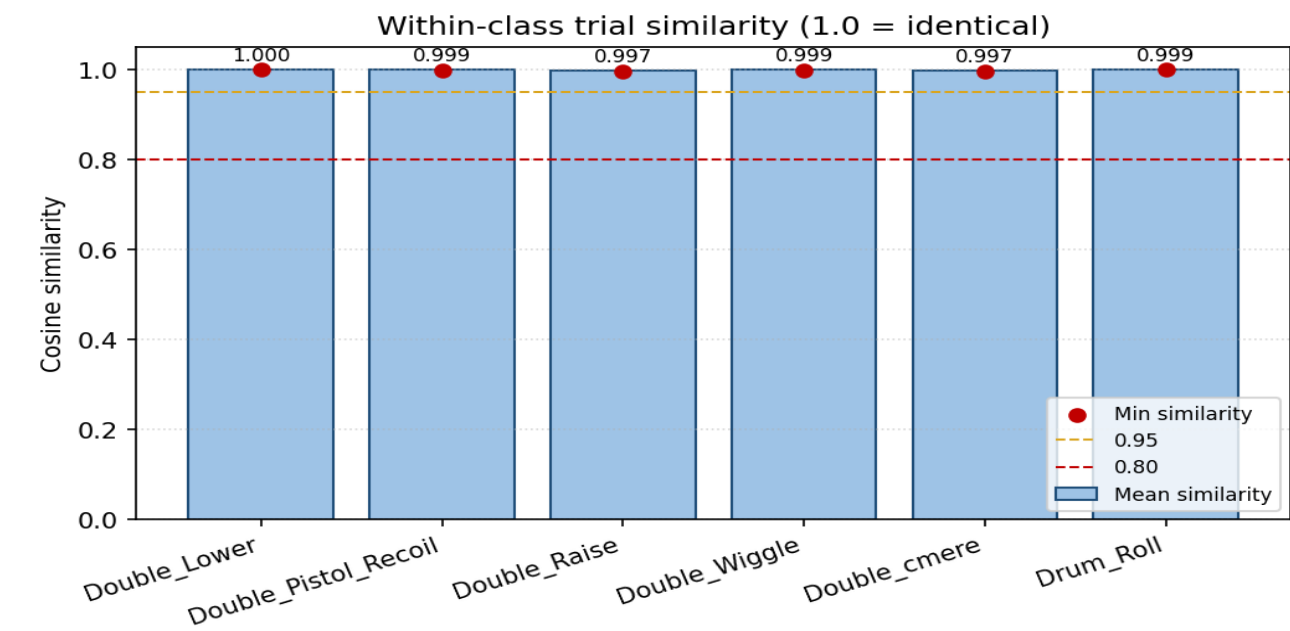


t-SNE feature space projection

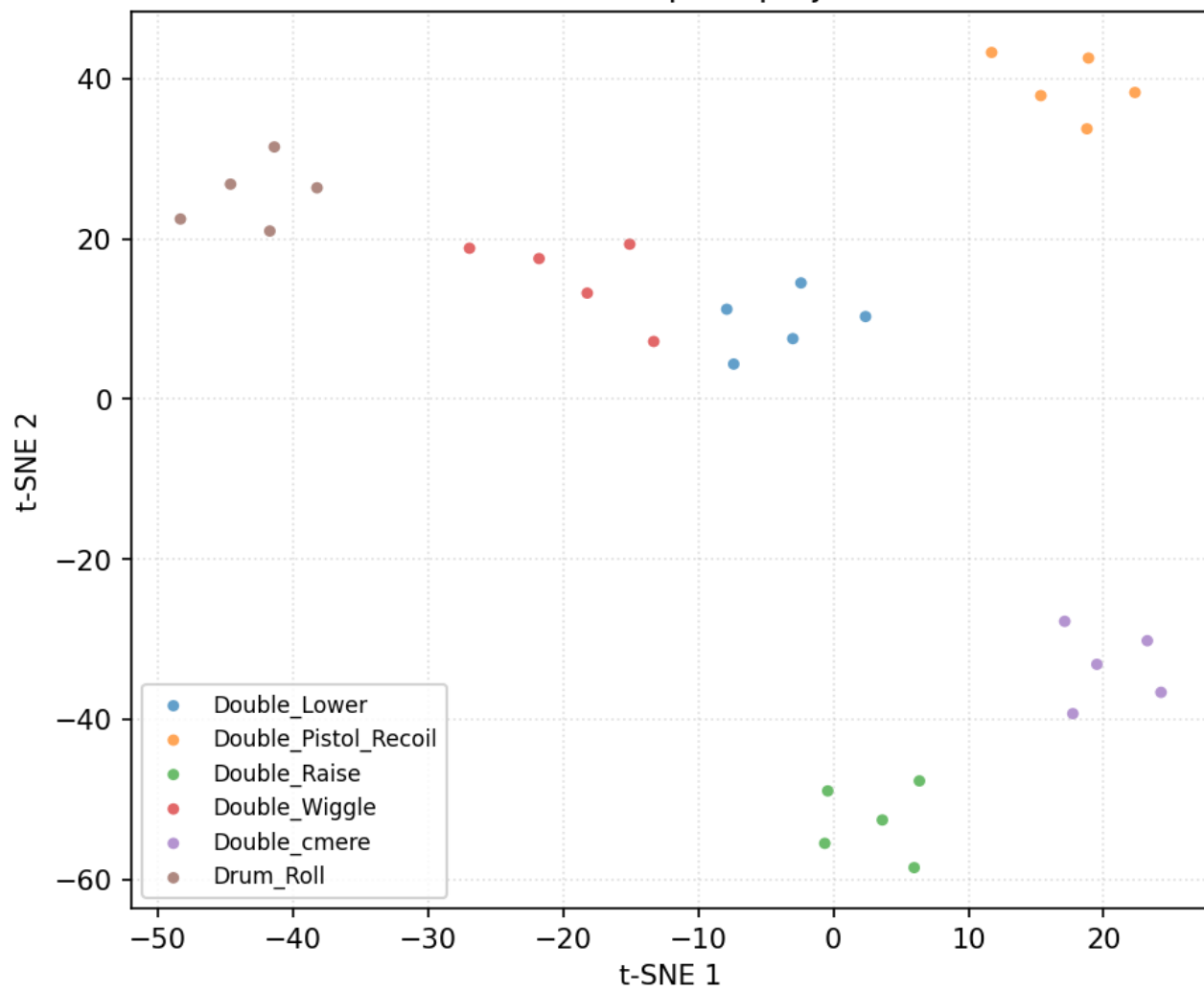


Data diagnostics — user 4_Daniel

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

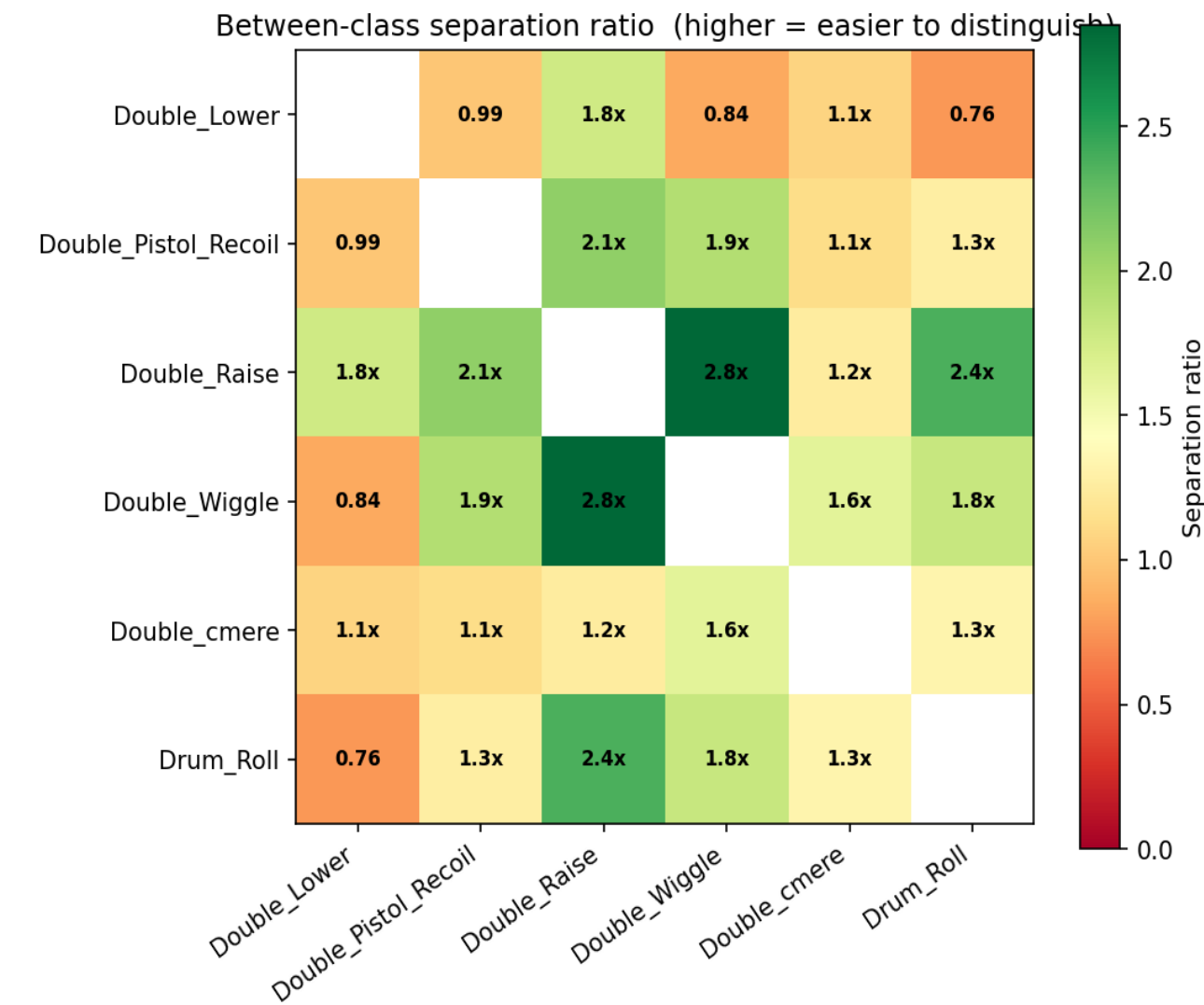
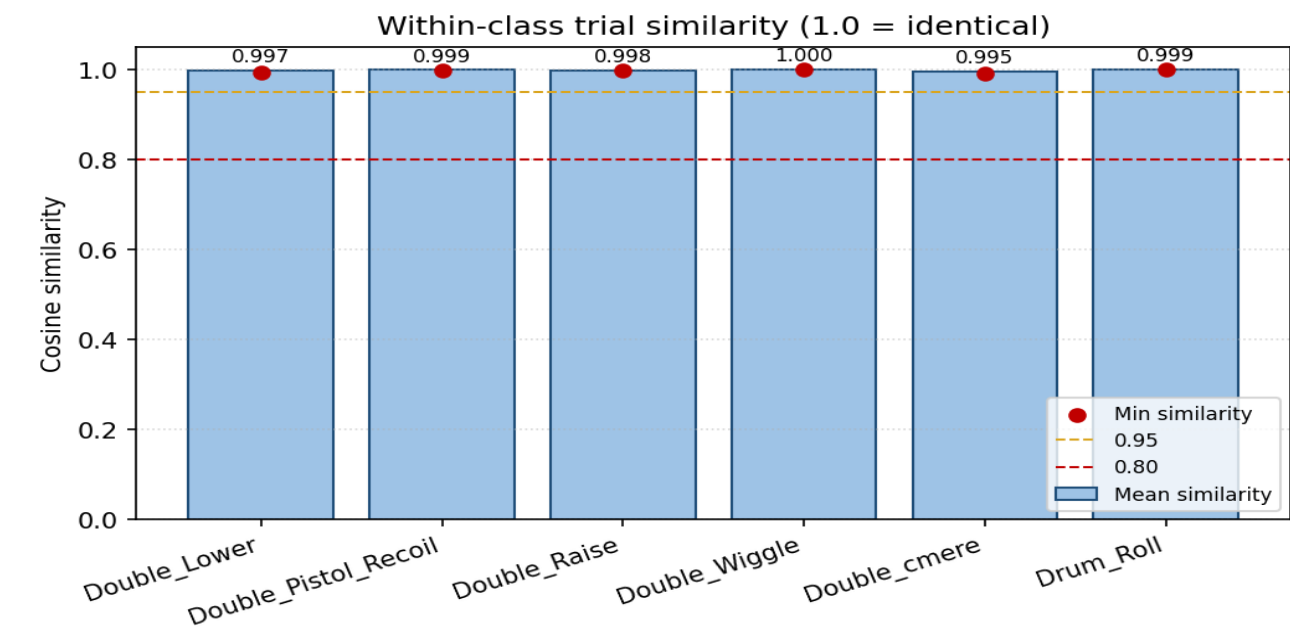


t-SNE feature space projection

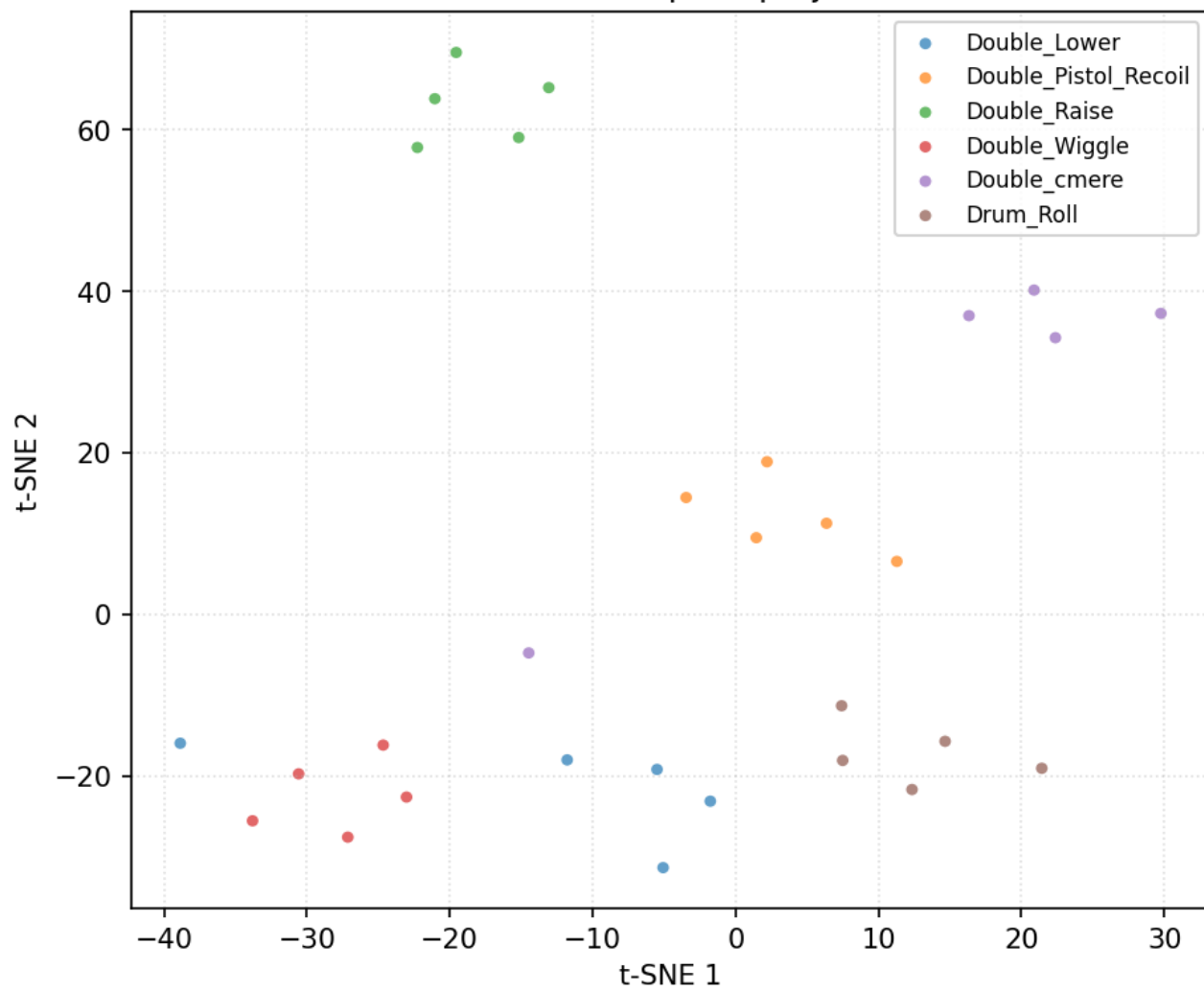


Data diagnostics — user 6_Harry

30 trials, 6 classes (filtered + resampled, before per-trial normalisation).

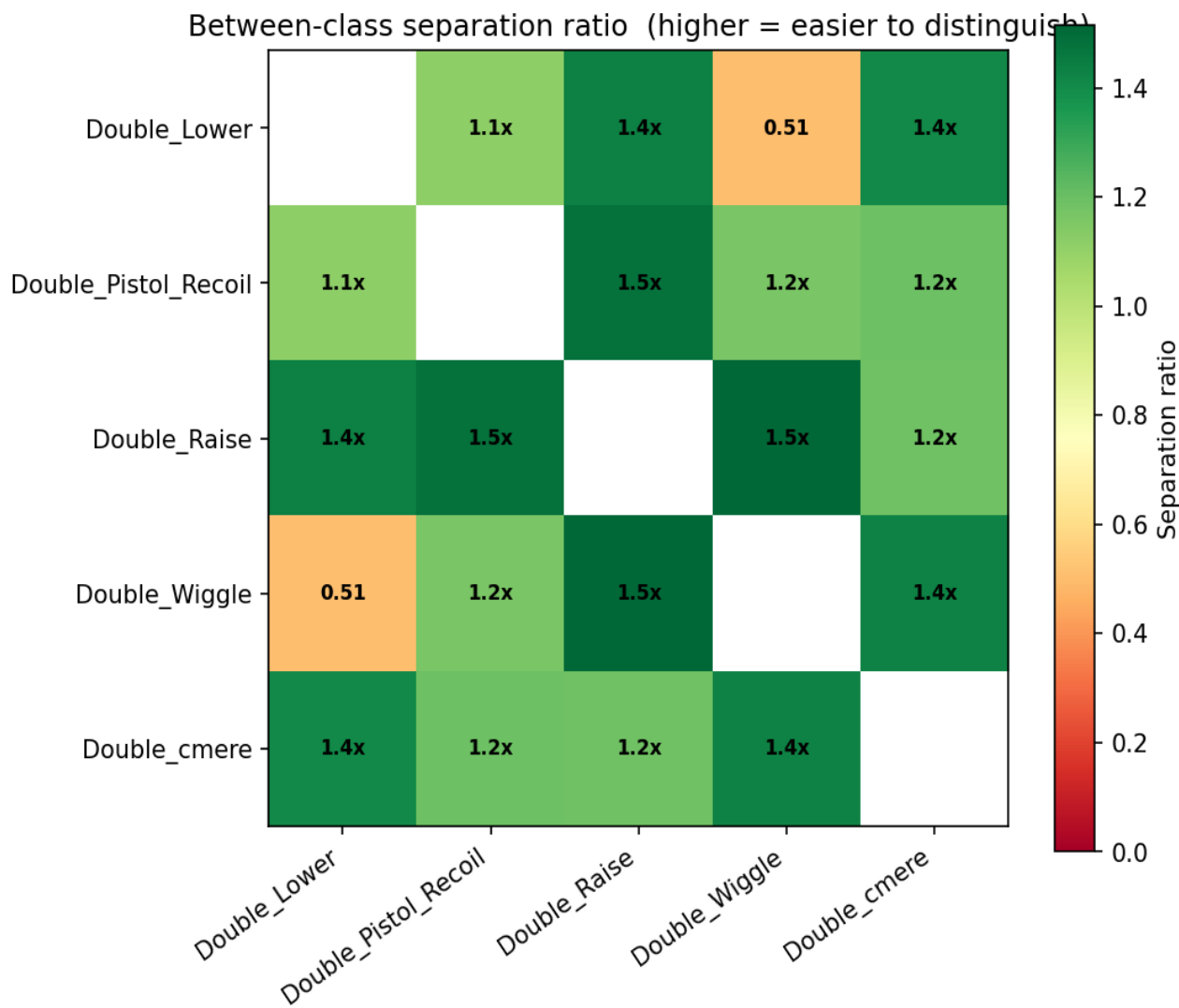
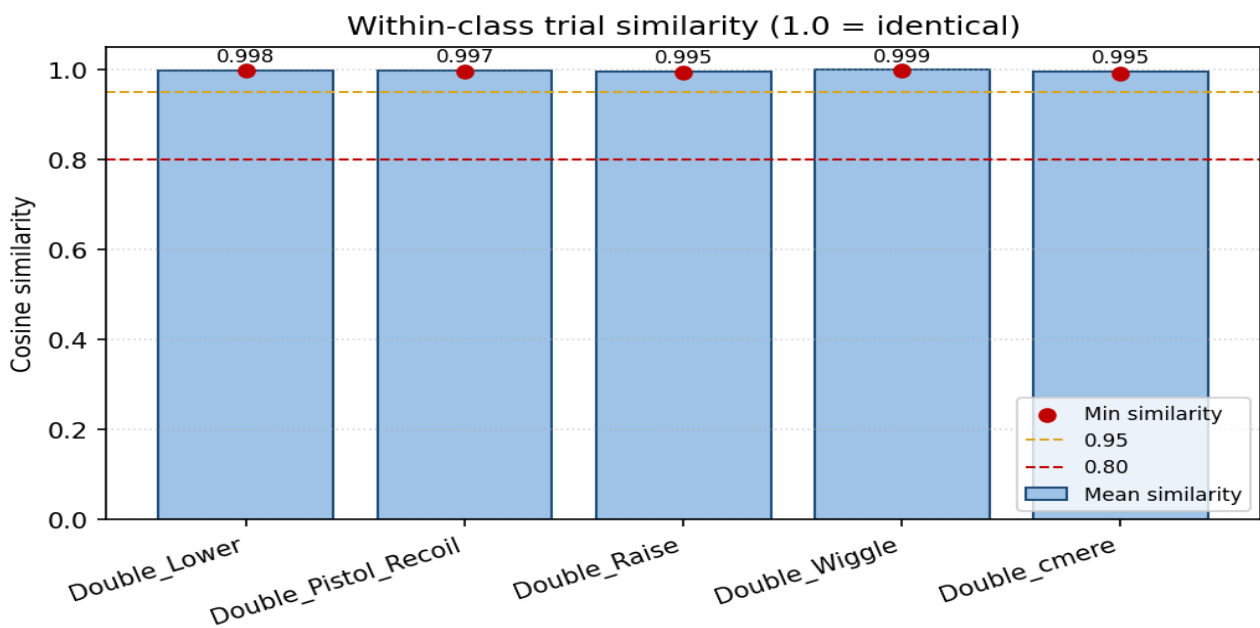


t-SNE feature space projection

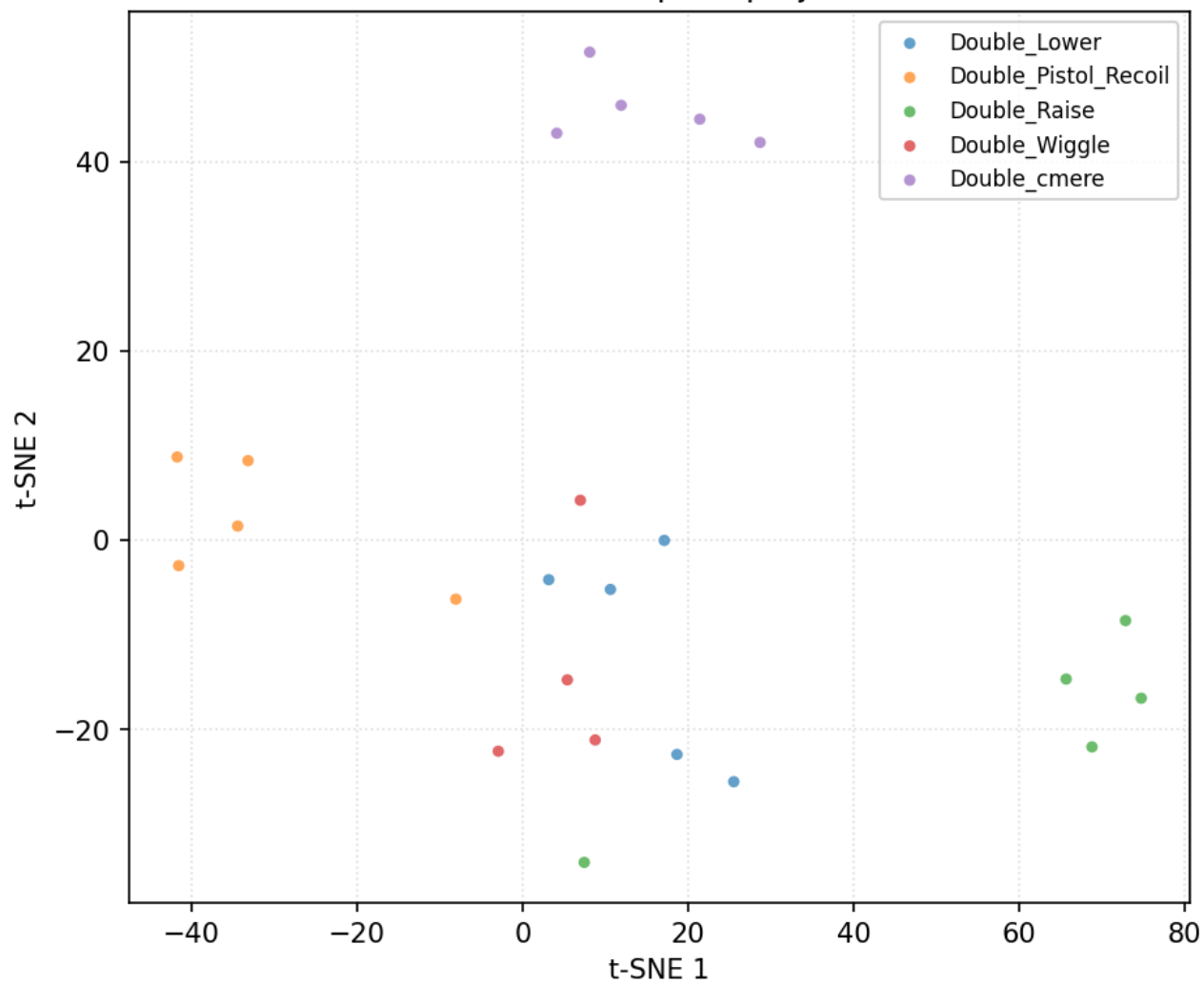


Data diagnostics — user 8_Jestin

24 trials, 5 classes (filtered + resampled, before per-trial normalisation).



t-SNE feature space projection



Variant: CNN_BiLSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→BiLSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	105,510
LOUO mean ± std	0.9821 ± 0.0249
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

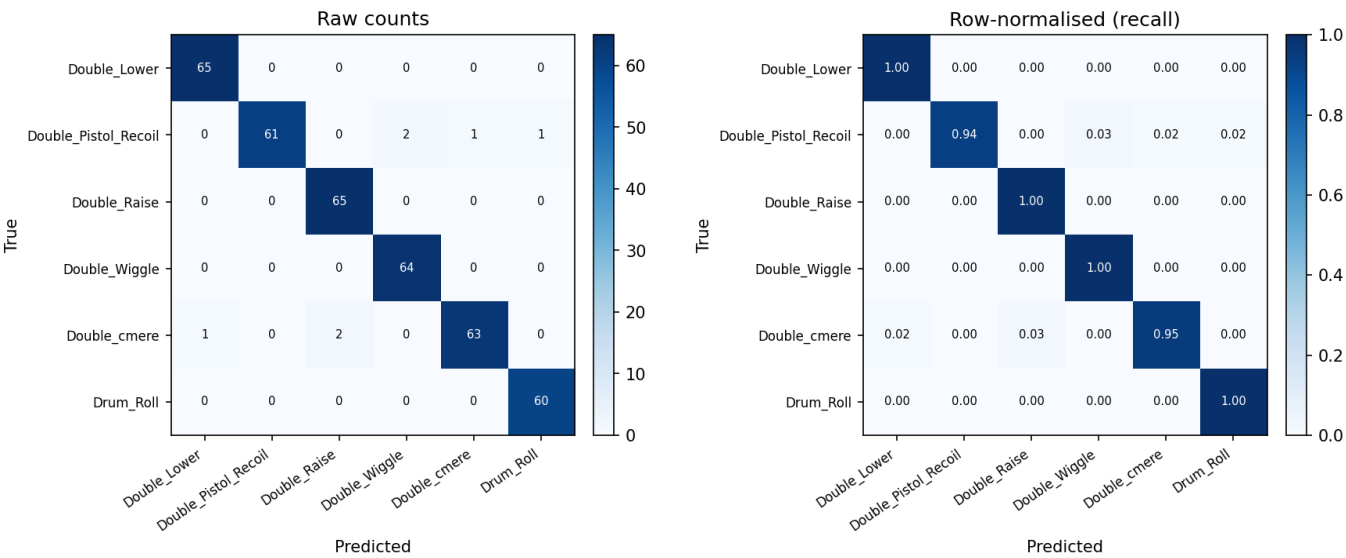
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	325	30	1.0000
2	12_Marcus	14_StephenV2	325	30	1.0000
3	14_StephenV2	15_Tash	325	30	0.9667
4	15_Tash	16_Bella	325	30	1.0000
5	16_Bella	17_Raquel	325	30	1.0000
6	17_Raquel	18_Bridgette	325	30	1.0000
7	18_Bridgette	1_Alan	325	30	0.9667
8	1_Alan	2_Alex	325	30	0.9333
9	2_Alex	3_Anghad	324	30	0.9667
10	3_Anghad	4_Daniel	324	31	1.0000
11	4_Daniel	6_Harry	325	30	1.0000
12	6_Harry	8_Jestin	331	30	0.9333
13	8_Jestin	11_Mansh	331	24	1.0000

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.985	1.000	0.992	65
Double_Pistol_Recoil	1.000	0.938	0.968	65
Double_Raise	0.970	1.000	0.985	65
Double_Wiggle	0.970	1.000	0.985	64
Double_cmere	0.984	0.955	0.969	66
Drum_Roll	0.984	1.000	0.992	60
macro avg	0.982	0.982	0.982	385
weighted avg	0.982	0.982	0.982	385

Confusion matrix (aggregated across folds)



Variant: BN_GAP

Conv+BN+ReLU ×3 with GlobalAveragePooling1D head — fewer params, less overfit

Trainable params	57,190
LOUO mean ± std	0.9795 ± 0.0463
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

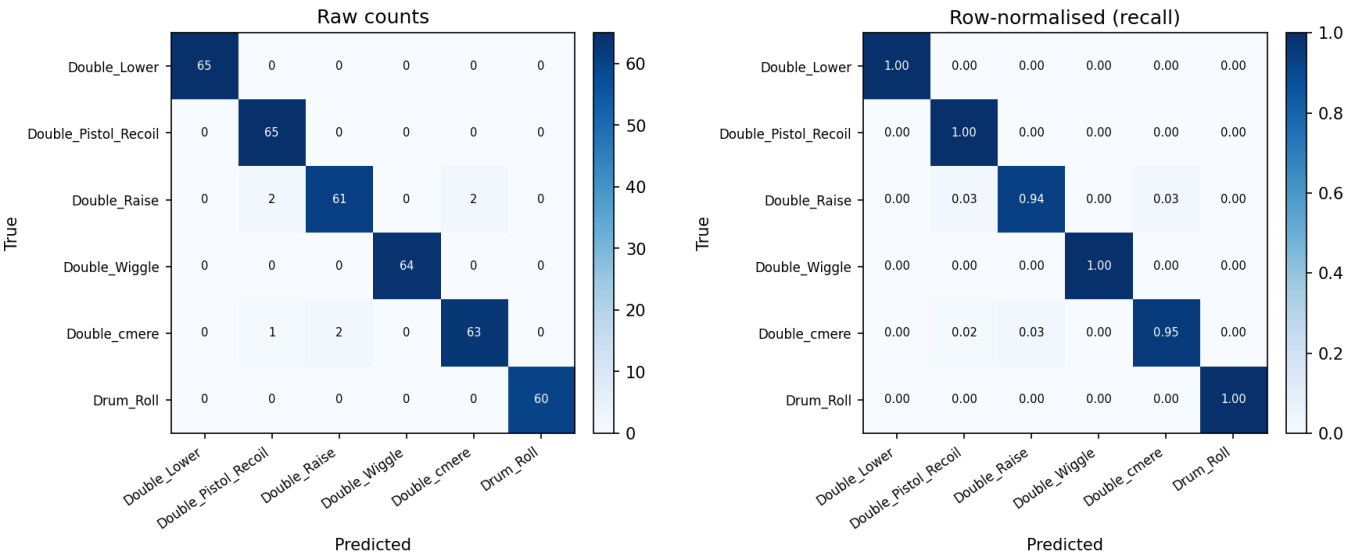
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
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2	12_Marcus	14_StephenV2	325	30	1.0000
3	14_StephenV2	15_Tash	325	30	1.0000
4	15_Tash	16_Bella	325	30	1.0000
5	16_Bella	17_Raquel	325	30	1.0000
6	17_Raquel	18_Bridgette	325	30	1.0000
7	18_Bridgette	1_Alan	325	30	1.0000
8	1_Alan	2_Alex	325	30	0.9333
9	2_Alex	3_Anghad	324	30	1.0000
10	3_Anghad	4_Daniel	324	31	1.0000
11	4_Daniel	6_Harry	325	30	1.0000
12	6_Harry	8_Jestin	331	30	0.9667
13	8_Jestin	11_Mansh	331	24	0.8333

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	1.000	1.000	1.000	65
Double_Pistol_Recoil	0.956	1.000	0.977	65
Double_Raise	0.968	0.938	0.953	65
Double_Wiggle	1.000	1.000	1.000	64
Double_cmere	0.969	0.955	0.962	66
Drum_Roll	1.000	1.000	1.000	60
macro avg	0.982	0.982	0.982	385
weighted avg	0.982	0.982	0.982	385

Confusion matrix (aggregated across folds)



Variant: Baseline

Conv1D(32,k=4)→Pool→Conv1D(64,k=4)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	51,750
LOUO mean ± std	0.9718 ± 0.0450
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

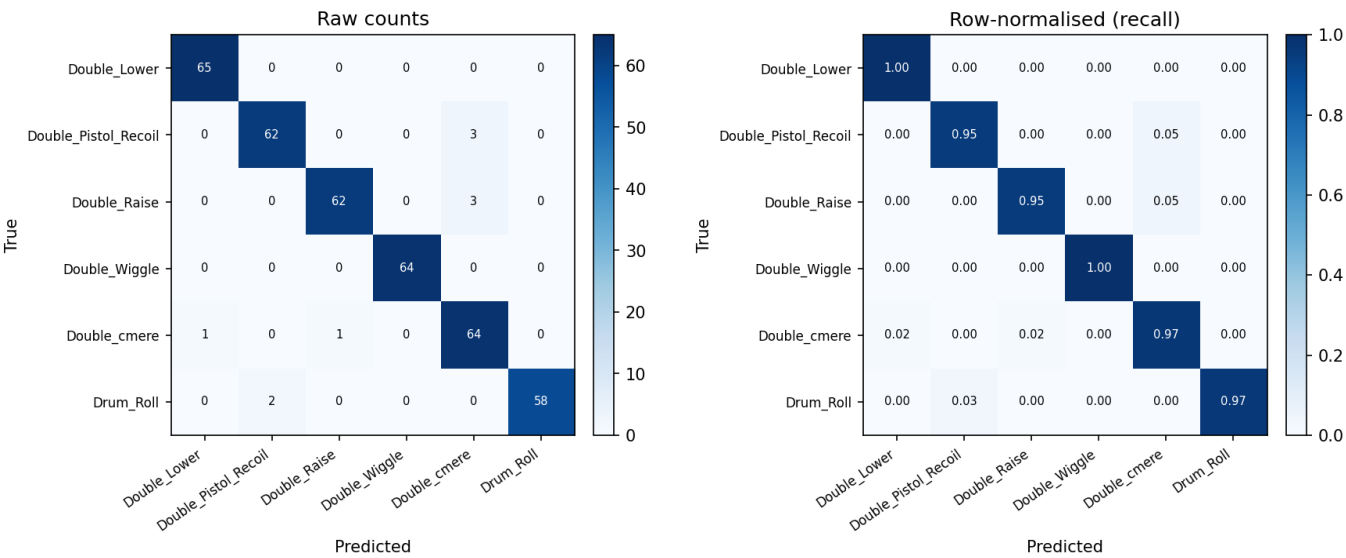
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5	16_Bella	17_Raquel	325	30	1.0000
6	17_Raquel	18_Bridgette	325	30	1.0000
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10	3_Anghad	4_Daniel	324	31	1.0000
11	4_Daniel	6_Harry	325	30	1.0000
12	6_Harry	8_Jestin	331	30	0.9667
13	8_Jestin	11_Mansh	331	24	0.8333

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.985	1.000	0.992	65
Double_Pistol_Recoil	0.969	0.954	0.961	65
Double_Raise	0.984	0.954	0.969	65
Double_Wiggle	1.000	1.000	1.000	64
Double_cmere	0.914	0.970	0.941	66
Drum_Roll	1.000	0.967	0.983	60
macro avg	0.975	0.974	0.974	385
weighted avg	0.975	0.974	0.974	385

Confusion matrix (aggregated across folds)



Variant: Deep

Three Conv blocks: Conv1D(32,4)→Conv1D(64,4)→Conv1D(128,3)→Flatten→Dense(128)→Dropout(0.4)→Softmax

Trainable params	72,806
LOUO mean ± std	0.9699 ± 0.0386
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

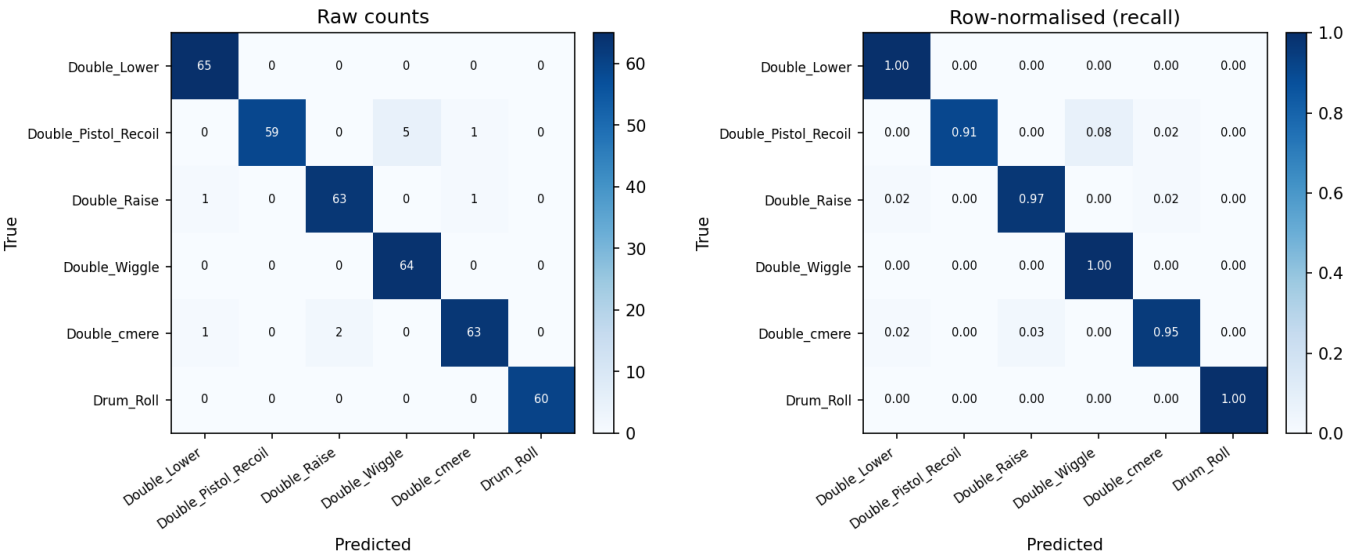
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	325	30	1.0000
2	12_Marcus	14_StephenV2	325	30	0.9667
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4	15_Tash	16_Bella	325	30	1.0000
5	16_Bella	17_Raquel	325	30	1.0000
6	17_Raquel	18_Bridgette	325	30	1.0000
7	18_Bridgette	1_Alan	325	30	0.9667
8	1_Alan	2_Alex	325	30	0.9000
9	2_Alex	3_Anghad	324	30	0.9667
10	3_Anghad	4_Daniel	324	31	1.0000
11	4_Daniel	6_Harry	325	30	1.0000
12	6_Harry	8_Jestin	331	30	0.9667
13	8_Jestin	11_Mansh	331	24	0.8750

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.970	1.000	0.985	65
Double_Pistol_Recoil	1.000	0.908	0.952	65
Double_Raise	0.969	0.969	0.969	65
Double_Wiggle	0.928	1.000	0.962	64
Double_cmere	0.969	0.955	0.962	66
Drum_Roll	1.000	1.000	1.000	60
macro avg	0.973	0.972	0.972	385
weighted avg	0.972	0.971	0.971	385

Confusion matrix (aggregated across folds)



Variant: CNN_LSTM

Conv1D(32,4)→Pool→Conv1D(64,4)→Pool→Dropout→LSTM(64)→Dense(64)→Dropout→Softmax

Trainable params	68,390
LOUO mean ± std	0.9686 ± 0.0581
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

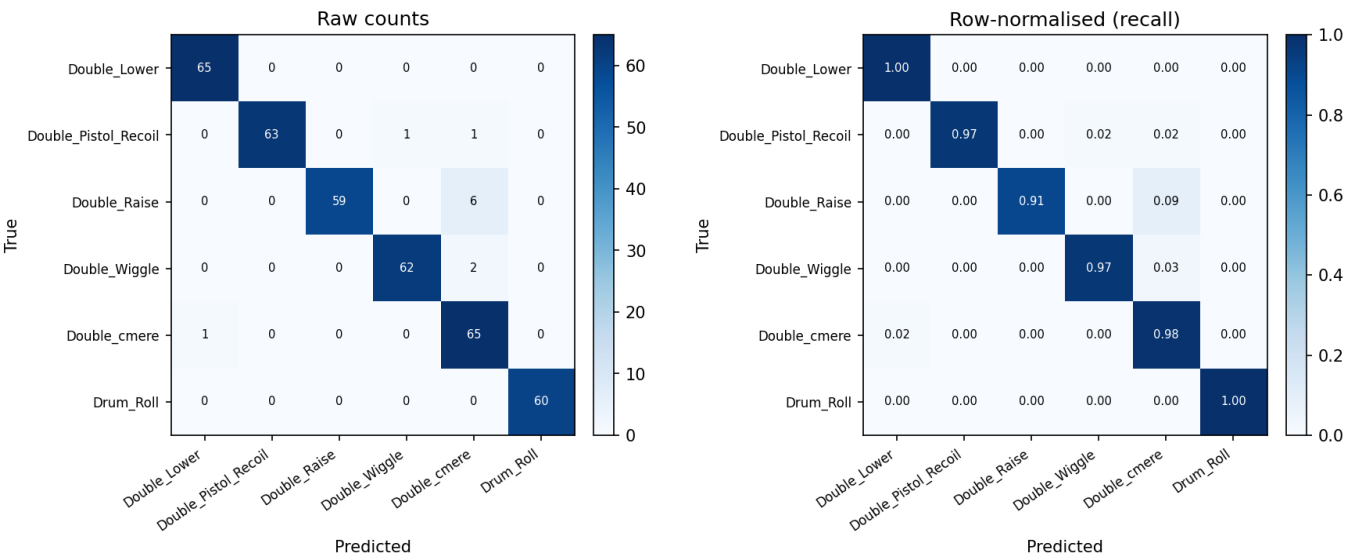
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	325	30	1.0000
2	12_Marcus	14_StephenV2	325	30	1.0000
3	14_StephenV2	15_Tash	325	30	1.0000
4	15_Tash	16_Bella	325	30	1.0000
5	16_Bella	17_Raquel	325	30	1.0000
6	17_Raquel	18_Bridgette	325	30	1.0000
7	18_Bridgette	1_Alan	325	30	0.9667
8	1_Alan	2_Alex	325	30	0.9000
9	2_Alex	3_Anghad	324	30	0.9667
10	3_Anghad	4_Daniel	324	31	1.0000
11	4_Daniel	6_Harry	325	30	1.0000
12	6_Harry	8_Jestin	331	30	0.9667
13	8_Jestin	11_Mansh	331	24	0.7917

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.985	1.000	0.992	65
Double_Pistol_Recoil	1.000	0.969	0.984	65
Double_Raise	1.000	0.908	0.952	65
Double_Wiggle	0.984	0.969	0.976	64
Double_cmere	0.878	0.985	0.929	66
Drum_Roll	1.000	1.000	1.000	60
macro avg	0.975	0.972	0.972	385
weighted avg	0.974	0.971	0.972	385

Confusion matrix (aggregated across folds)



Variant: WideKernel

Conv1D(32,k=8)→Pool→Conv1D(64,k=8)→Pool→Flatten→Dense(64)→Dropout(0.3)→Softmax

Trainable params	70,182
LOUO mean ± std	0.9661 ± 0.0321
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

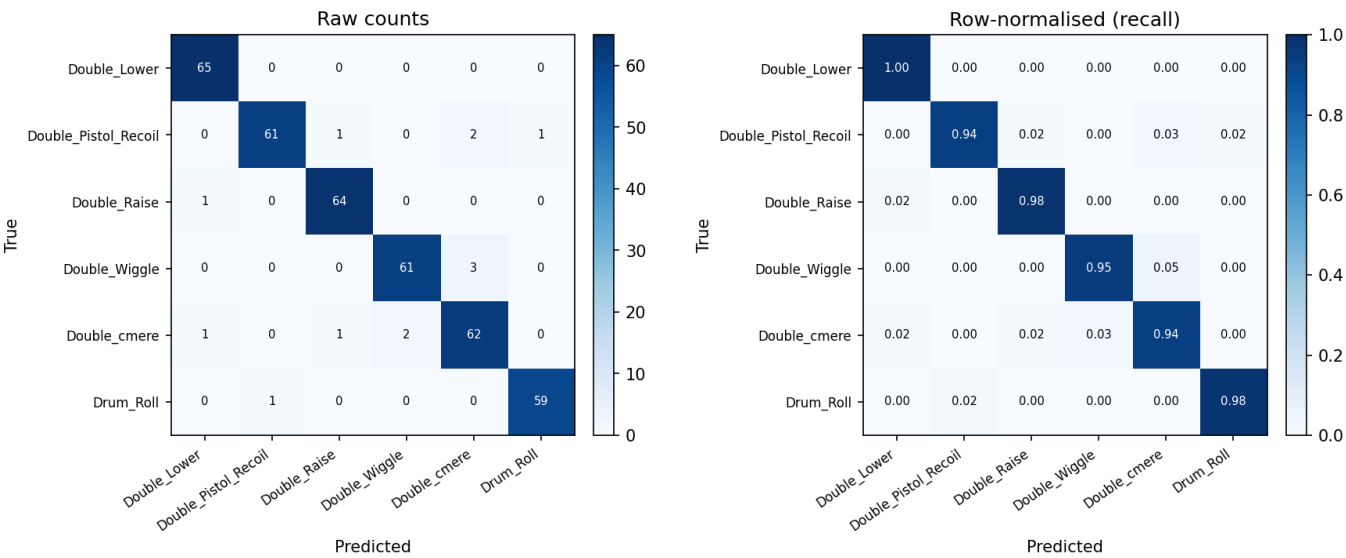
LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	325	30	1.0000
2	12_Marcus	14_StephenV2	325	30	1.0000
3	14_StephenV2	15_Tash	325	30	0.9667
4	15_Tash	16_Bella	325	30	0.9333
5	16_Bella	17_Raquel	325	30	1.0000
6	17_Raquel	18_Bridgette	325	30	1.0000
7	18_Bridgette	1_Alan	325	30	0.9333
8	1_Alan	2_Alex	325	30	0.9667
9	2_Alex	3_Anghad	324	30	0.9333
10	3_Anghad	4_Daniel	324	31	0.9677
11	4_Daniel	6_Harry	325	30	1.0000
12	6_Harry	8_Jestin	331	30	0.9000
13	8_Jestin	11_Mansh	331	24	0.9583

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.970	1.000	0.985	65
Double_Pistol_Recoil	0.984	0.938	0.961	65
Double_Raise	0.970	0.985	0.977	65
Double_Wiggle	0.968	0.953	0.961	64
Double_cmere	0.925	0.939	0.932	66
Drum_Roll	0.983	0.983	0.983	60
macro avg	0.967	0.966	0.966	385
weighted avg	0.966	0.966	0.966	385

Confusion matrix (aggregated across folds)



Variant: Shallow

Single Conv block: Conv1D(32,k=5)→Pool→Flatten→Dense(32)→Dropout(0.3)→Softmax

Trainable params	41,734
LOUO mean ± std	0.9635 ± 0.0402
Folds run	13
Inner-val mode	user
Per-trial norm	zscore

LOUO per-fold results

Fold	Held-out user	Inner-val user	n_train	n_test	Accuracy
1	11_Mansh	12_Marcus	325	30	1.0000
2	12_Marcus	14_StephenV2	325	30	0.9000
3	14_StephenV2	15_Tash	325	30	0.9667
4	15_Tash	16_Bella	325	30	1.0000
5	16_Bella	17_Raquel	325	30	1.0000
6	17_Raquel	18_Bridgette	325	30	1.0000
7	18_Bridgette	1_Alan	325	30	1.0000
8	1_Alan	2_Alex	325	30	0.8667
9	2_Alex	3_Anghad	324	30	0.9667
10	3_Anghad	4_Daniel	324	31	0.9677
11	4_Daniel	6_Harry	325	30	0.9667
12	6_Harry	8_Jestin	331	30	0.9333
13	8_Jestin	11_Mansh	331	24	0.9583

Per-class metrics (aggregated across folds)

Class	Precision	Recall	F1	Support
Double_Lower	0.970	1.000	0.985	65
Double_Pistol_Recoil	0.984	0.938	0.961	65
Double_Raise	0.955	0.969	0.962	65
Double_Wiggle	0.967	0.922	0.944	64
Double_cmere	0.984	0.955	0.969	66
Drum_Roll	0.923	1.000	0.960	60
macro avg	0.964	0.964	0.963	385
weighted avg	0.964	0.964	0.964	385

Confusion matrix (aggregated across folds)

